ColRadPy

Theory Manual

Collisional radiative kinetics and spectral emission

Code version	1.1 (`pyproject.toml`)
Source branch	`development`
Source record	HEAD and file hashes in `provenance.json`
Code authors	C. A. Johnson and contributors (original code)
	G. Ronchi (`development` branch)
Reference	C. A. Johnson, S. D. Loch and D. A. Ennis (2019)
Manual date	October 2026

This manual develops the rate equations, the elimination of excited states, generalized collisional radiative coefficients, line emission, time-dependent and charge-state solutions, and the energy-balance extension. Equations are mapped to the routines that implement them. Blue notes explain implementation choices; orange notes identify current limitations or qualifications. The worked checks execute the development working tree; file hashes and the recorded HEAD are given in `docs/source/theory/figures/provenance.json`.

Contents

1 Physical framework and conventions **1**
1.1 Kinetic descriptions . . . 1
1.2 Calculation flow . . . 1
1.3 Notation, indexing and units . . . 1

2 Atomic processes and rate coefficients **3**
2.1 Level structure and statistical weights . . . 3
2.2 Maxwellian electron-impact excitation . . . 3
2.3 Spontaneous decay, ionization and recombination . . . 4
2.4 Three-body recombination and Saha detailed balance . . . 5
2.5 Thermal charge exchange . . . 5

3 Level-population equations and their solutions **5**
3.1 Time-dependent evolution and matrix exponentials . . . 6
3.2 Stationary solutions . . . 6
3.3 A two-level density scan . . . 7
3.4 A conservative chain and a slow mode . . . 7

4 Quasistatic reduction and population normalization **8**
4.1 Choosing retained levels and testing time-scale separation . . . 9
4.2 Separate excitation, recombination and CX drivers . . . 10
4.3 Normalization is a choice of driving population . . . 10
4.4 Retained fractions and total ion fractions . . . 11

5 Effective coefficients and charge-state evolution **12**
5.1 SCD: effective ionization . . . 12
5.2 ACD and CCD: effective capture . . . 12
5.3 QCD: redistribution within a retained bound manifold . . . 13
5.4 XCD: redistribution between parent metastables . . . 13
5.5 XCD checked by direct elimination . . . 13
5.6 Ionization balance across charge stages . . . 13
5.6.1 Matrix assembly and density factors . . . 14
5.6.2 Ionization age . . . 14
5.7 Stage-to-stage coefficients . . . 14
5.7.1 Resolved metastables and ground condensation . . . 14
5.7.2 Contraction with metastable fractions . . . 15
5.7.3 When capture feeding changes the fractions . . . 15

6 Emission, power and spectroscopic diagnostics **16**
6.1 Photon emissivity coefficients . . . 16
6.2 Two-photon decay and the limits of line outputs . . . 16
6.3 Radiated power coefficients . . . 17
6.4 S/XB: ionizations per emitted photon . . . 18
6.5 Unsummed contributions and dominant feeding mechanisms . . . 18
6.6 Wavelengths, fine structure and line profiles . . . 19
6.6.1 Wavelengths and NIST energy matching . . . 19
6.6.2 Statistical splitting of LS multiplets . . . 20
6.6.3 Doppler and natural profiles . . . 20
6.7 Electron energy loss, radiated power and emission functions . . . 20

7 Projection of the high n-shells **21**
7.1 Condensed shell model and elimination . . . 21
7.2 Removing the direct processes . . . 22
7.3 Expansion onto the explicit levels . . . 22
7.4 Grid continuation and numerical qualifications . . . 23

8 Atomic-data preparation **23**
8.1 Interpolation and asymptotic continuation . . . 23
8.2 Atomic data from the Flexible Atomic Code . . . 25
8.2.1 Common thermal average . . . 25
8.2.2 Excitation and ionization . . . 25
8.2.3 Radiative recombination . . . 26
8.2.4 Dielectronic capture and autoionization folding . . . 26

9 Charge-state kinetic-energy evolution **27**
9.1 Charged and neutral energy transfer . . . 28

10 Applicability and verification **28**
10.1 Current limitations . . . 28
10.2 Analytic examples and checks . . . 30

A Data products and external interfaces **30**
A.1 Array layout . . . 30
A.2 ADF04, ADF11, ADF13, ADF15 and HDF5 . . . 30
A.3 Input routes and retained manifolds . . . 31

B Numerical methods for matrix evolution **32**
B.1 Constant sources and modal evaluation . . . 32
B.2 Complex modes and real populations . . . 32
B.3 Defective and ill-conditioned eigenbases . . . 32
B.4 Stationary numerical methods . . . 32

C How ADAS makes its ADF11 data **33**
C.1 Production of GCR-project coefficients . . . 33
C.2 XCD in the inspected ADAS208 implementation . . . 33
C.3 Resolved and unresolved charge-state data . . . 34
C.4 Resolved and unresolved emission data . . . 34

D Source-code map **34**

E Evidence and reproduction paths **36**

1 Physical framework and conventions

ColRadPy describes the populations and emission of atomic ions using a generalized collisional-radiative (GCR) model [1, 9]. Level-resolved atomic data determine the rates of excitation, decay, ionization and capture. The resulting kinetic equations yield both the populations of individual levels and effective coefficients for the evolution of charge states. An ADAS ADF04 file provides the usual atomic input, including level energies, radiative probabilities, effective collision strengths and any available parent-resolved ionization, recombination or charge-exchange data.

The basic model assumes a prescribed Maxwellian electron bath and optically thin emission. Temperature and density are external parameters, and emitted photons escape without further interaction. Ground and metastable populations may remain independent dynamical variables, while shorter-lived excited populations can be represented by their response to these variables. This local kinetic description supplies coefficients to transport and plasma models; radiative transfer, spatial transport and the electron distribution function are outside its scope.

The competition between radiative and collisional rates connects the coronal and high-density limits. At low density, spontaneous decay dominates the removal of ordinary excited populations. As density increases, reversible collisions can establish Boltzmann ratios within a charge stage. Ionization and three-body recombination approach Saha balance when the inverse rates and statistical weights are consistent. These limiting behaviors provide physical checks on a calculation. Implementation statements in this manual refer to the source recorded with the accompanying figures and numerical checks [12].

1.1 Kinetic descriptions

The level equations admit a full time-dependent solution and a quasistatic reduction. The former follows every supplied population from an initial state. The latter assumes that selected excited levels relax rapidly and expresses their populations in terms of retained ground, metastable and parent populations. Eliminating the excited levels also produces effective rates, which couple adjacent ions in a charge-state evolution model. The interpretation of each result follows from its driving populations and normalization.

The development below proceeds from atomic rates to level kinetics, quasistatic reduction, effective charge-state coefficients and emission. The two-level example introduces the role of density and population normalization before the general reduction. High-shell projection, atomic-data preparation and kinetic-energy evolution extend this framework in Sections 7–9. The final section addresses applicability and verification; the appendices document numerical algorithms, data conventions, the ADAS production chain and the source and evidence supporting the manual.

1.2 Calculation flow

1.3 Notation, indexing and units

Let $N_i^{(z)}$ denote the number density of level i in charge state z. Bound levels are indexed by i, j, retained ground and metastable levels by ρ, σ, and parents in charge state $z+1$ by p, q. Matrix-block subscripts identify sets of states: e denotes eliminated excited levels, m retained bound levels, and r all retained states, including the parents. Electron density is written explicitly as n_e.

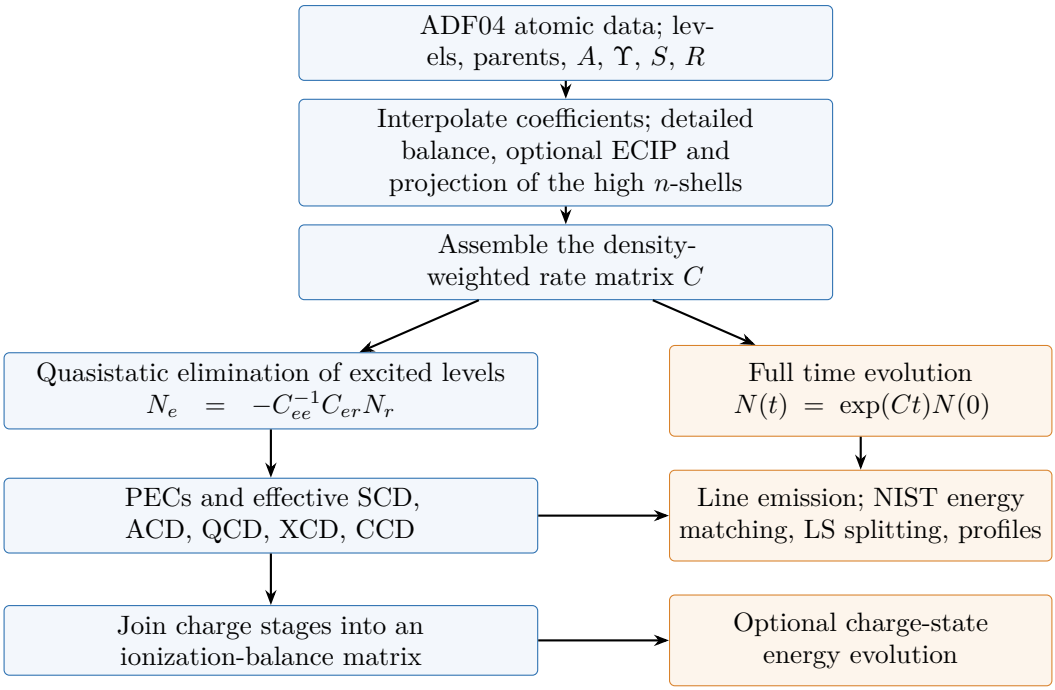


Figure 1: The calculation paths are related but do not share the same population normalization. The full matrix can also include parent or reservoir states; their interpretation must be specified.

The defining matrix convention is

$$\frac{\mathrm{d}N_i}{\mathrm{d}t} = \sum_j C_{ij}N_j, \qquad C_{ij}: \; j \longrightarrow i. \tag{1.1}$$

Each column therefore represents a source state and each row a destination. Physical transfer rates give nonnegative off-diagonal entries. Some stored coefficient arrays use a different index order, documented in Appendix A.1.

Quantity	Meaning	Units
T_e, T_i	Temperature represented as an energy	eV
$E_i,\ I_p$	Tabulated level energies and parent thresholds	cm^{-1}
$n_\mathrm{e}, n_\mathrm{h}, N_i$	Electron, neutral-donor and level densities	cm^{-3}
$A_{u\ell}$	Spontaneous emission probability	s^{-1}
$q, S, \alpha^{\mathrm{RR+DR}}, K$	Two-body coefficients	$\mathrm{cm}^3\,\mathrm{s}^{-1}$
$\alpha^{(3)}$	Three-body recombination coefficient	$\mathrm{cm}^6\,\mathrm{s}^{-1}$
C	Density-weighted rate matrix	s^{-1}
Υ	Effective collision strength	dimensionless
λ	Internal wavelength	nm
PEC	Photon emission coefficient	$\mathrm{cm}^3\,\mathrm{s}^{-1}$
SCD, ACD, QCD, XCD, CCD	Effective two-body-form coefficients	$\mathrm{cm}^3\,\mathrm{s}^{-1}$

Table 1: Core conventions. Effective coefficients may themselves depend on density. The energy-transfer helper functions use SI densities and masses.

The conversions needed repeatedly are

$$\Delta E_{\mathrm{eV}} = \frac{100\,hc}{e}\,\Delta E_{\mathrm{cm}^{-1}} \simeq \frac{\Delta E_{\mathrm{cm}^{-1}}}{8065.544}, \qquad T_{\mathrm{K}} = \frac{e}{k_B}\,T_{\mathrm{eV}} \simeq 11604.518\,T_{\mathrm{eV}}, \tag{1.2}$$

$$\lambda_{\mathrm{vac,nm}} = \frac{10^7}{|E_u - E_\ell|_{\mathrm{cm}^{-1}}}, \qquad n_{\mathrm{m}^{-3}} = 10^6 n_{\mathrm{cm}^{-3}}. \tag{1.3}$$

The energy and temperature conversions use the CODATA values supplied by `scipy.constants`; the factor 100 converts wavenumbers from cm^{-1} to m^{-1}. Several ADAS-derived rate expressions retain their original rounded constants: 13.6058 eV and 2.1716×10^{-8} in Eq. (2.2), 157890 K per rydberg in ECIP, and 1.656742×10^{-22} in Eq. (2.7). Numerical reproduction should follow the constants used by the relevant routine.

Unless specified as tabulated wavenumbers, ΔE, χ and FAC energies in rate expressions are in eV. Consequently, ratios such as $\Delta E/T_{\mathrm{e}}$ are dimensionless. Expressions involving SI constants require energies in joules, with $k_B T_{\mathrm{K}} = eT_{\mathrm{e}}$. Photon energies in power and electron-energy-loss coefficients are likewise in joules. The wavelength conversion uses the tabulated wavenumber differences.

ADF04 level and parent identifiers are one-based, whereas Python indices are zero-based. Thus `metas=[0,2]` retains the first and third bound levels, leaving $[1, 3, 4, \ldots]$ as the eliminated set. This explicit complement is needed when the retained levels are noncontiguous. Parent identifiers must also be distinguished from their discovery order in the input file.

2 Atomic processes and rate coefficients

2.1 Level structure and statistical weights

Statistical weights reflect the resolution of the atomic model. An LS term has weight $(2S + 1)(2L + 1)$, while a resolved-J level has weight $2J + 1$. ColRadPy represents this through the ADF04 field `atomic['w']` and uses $g_i = 2w_i + 1$ in rate construction. Parent weights must follow the same resolution. Parent ionization thresholds are specified in the file header, while level annotations identify parentage channels; these two records need not enumerate the same observed pathways [3].

Parent weights. `parent_statistical_weight` interprets LS and resolved-J parent labels, including fractional J, to obtain $(2S + 1)(2L + 1)$ or $2J + 1$. Explicit ionization data may be present even when the associated parentage annotations are incomplete, so channel identifiers and header counts require consistent mapping.

2.2 Maxwellian electron-impact excitation

Electron-impact coefficients are averages of the cross-section over the electron distribution. For a normalized energy probability density f_E in eV^{-1}, a cross-section σ in cm^2 and electron speed v in cm/s, the excitation coefficient is

$$q_{\ell u} = \int_{\Delta E}^{\infty} \sigma_{\ell u}(E) v(E)\, f_E(E)\, \mathrm{d}E. \tag{2.1}$$

Both the incident energy E and threshold ΔE are in eV, giving q in cm^3/s. ADF04 expresses the nonrelativistic Maxwellian average through the dimensionless effective collision strength $\Upsilon_{\ell u}(T)$.

For $u > \ell$, the downward coefficient and its upward inverse are

$$q_{u\ell}(T_\mathrm{e}) = \frac{2.1716 \times 10^{-8}}{g_u} \sqrt{\frac{13.6058}{T_\mathrm{e}}}\, \Upsilon_{\ell u}(T_\mathrm{e}), \tag{2.2}$$

$$q_{\ell u}(T_\mathrm{e}) = \frac{g_u}{g_\ell} \exp\left(-\frac{\Delta E}{T_\mathrm{e}}\right) q_{u\ell}(T_\mathrm{e}). \tag{2.3}$$

The Boltzmann factor uses ΔE and T_e in the same energy units. The ratio of these coefficients gives $N_u/N_\ell = (g_u/g_\ell)\exp(-\Delta E/T_\mathrm{e})$ when collisions control the level ratio. This detailed-balance relation is the basis of the within-stage Boltzmann limit. Section 8.2 describes the corresponding treatment of cross-section data from FAC.

Implementation. make_electron_excitation_rates interpolates Υ to the requested temperatures. populate_cr_matrix constructs the upward and downward coefficients and applies the electron-density factor. Boltzmann factors for excitation, ionization and three-body recombination use the wavenumber-to-eV conversion in Eq. (1.2); ECIP retains the constants of r8ecip.

2.3 Spontaneous decay, ionization and recombination

Spontaneous decay transfers population from u to ℓ at rate $A_{u\ell}N_u$, independently of electron density. Summation over the accessible lower levels gives the total radiative loss rate, $\Gamma_u = \sum_{\ell<u} A_{u\ell}$.

Ionization connects a bound level i to a parent p of the next charge stage. In the supported ADF04 convention, the tabulated reduced coefficient $\widetilde{S}_{ip}$ is related to the physical coefficient by

$$S_{ip}(T_\mathrm{e}) = \widetilde{S}_{ip}(T_\mathrm{e}) \exp[-\chi_{ip}/T_\mathrm{e}], \qquad \chi_{ip} = \frac{100\,hc}{e}\,(I_p - E_i) \quad (\mathrm{eV}). \tag{2.4}$$

The threshold factor is applied after interpolation of $\widetilde{S}_{ip}$. The option `scale_file_ioniz` additionally applies the input parentage factor to the interpolated reduced coefficients, independently of the interpolation method, and is passed to each stage by the ionization-balance calculation. Suspected low-temperature placeholders are filtered before interpolation because they would otherwise propagate into the inverse three-body rates. Section 8.1 specifies these heuristic filters and the warnings that identify affected channels and temperatures.

The file's radiative and dielectronic capture contributions are denoted by $\alpha_{ip}^{\mathrm{RR+DR}}$. Including three-body recombination gives the density-dependent coefficient

$$R_{ip}(T_\mathrm{e}, n_\mathrm{e}) = \alpha_{ip}^{\mathrm{RR+DR}}(T_\mathrm{e}) + n_\mathrm{e}\alpha_{ip}^{(3)}(T_\mathrm{e}). \tag{2.5}$$

The population-transfer rate is $n_\mathrm{e}R_{ip}N_p$. Radiative and dielectronic capture therefore enter linearly in electron density, whereas three-body capture enters quadratically. A zero or missing file entry alone does not establish that the corresponding physical process is absent.

Missing ionization channels may be supplemented with the approximate ECIP model. For each parent p, ecip_rates selects levels annotated $\{p\}$ below that parent's threshold and passes temperatures in kelvin to r8ecip. Levels above the first ionization threshold may thus contribute to higher-parent channels. suppliment_with_ecip uses parent annotations and zero entries in the assembled ionization array to select the supplemented channels. The resulting rates provide an approximate completion whose effect should be included in atomic-model convergence studies.

2.4 Three-body recombination and Saha detailed balance

Three-body recombination is related to ionization by equilibrium of $i + e \rightleftharpoons p + e + e$. Equality of the forward and inverse population-transfer rates requires

$$N_i^* n_\mathrm{e} S_{ip} = N_p^* n_\mathrm{e}^2 \alpha_{ip}^{(3)}. \tag{2.6}$$

For nondegenerate, nonrelativistic electrons, the translational partition factor then gives

$$\begin{aligned} \frac{\alpha_{ip}^{(3)}}{S_{ip}} &= \frac{g_i}{g_p} \frac{h^3}{2(2\pi m_e k_B T_\mathrm{K})^{3/2}} \exp\left(\frac{\chi_{ip}}{T_\mathrm{e}}\right) \\ &\simeq \frac{g_i}{g_p} \frac{1.6564 \times 10^{-22}}{T_\mathrm{e}^{3/2}} \exp\left(\frac{\chi_{ip}}{T_\mathrm{e}}\right) \ \mathrm{cm}^3. \end{aligned} \tag{2.7}$$

The numerical form uses T_e in eV, gives a volume in cm^3, and includes the free-electron spin factor of two. The implementation uses the nearby value 1.656742×10^{-22}. Evaluation of the first form with SI constants instead gives m^3, requiring a factor 10^6 for comparison. Its nonrelativistic basis is relevant when FAC forward rates use relativistic thermal averages (Section 8.2).

Stable low-temperature evaluation. The inverse rate is evaluated directly from reduced ionization data, avoiding multiplication of an underflowed ionization coefficient by an overflowing inverse exponential. For finite positive reduced data this produces a finite positive value at fixed positive temperature; the construction makes no assertion about a universal $T \to 0$ limit.

Standalone LTE helpers. boltzmann_population returns ground-relative level populations or total-normalized fractions. saha_population includes both stage partition functions, ionization energy, electron mass, and electron density; it returns N_{z+1}/N_z or the two-stage ionized fraction. Energies are in cm^{-1}, temperature in eV, and electron density in cm^{-3}.

2.5 Thermal charge exchange

Thermal charge exchange introduces a second bath: a donor species at temperature T_h and density n_h. Capture from parent p into level i contributes $n_\mathrm{h} K_{ip}(T_h) N_p^{\mathrm{CX}}$ to the bound-level balance. The rate is controlled by donor density. make_cx_rates_from_file interpolates the donor-temperature data; its grid must be compatible with the chosen electron-temperature/density grid or paired layout.

3 Level-population equations and their solutions

The level populations evolve under a rate matrix assembled from the atomic coefficients and the appropriate bath densities. In the source-column convention of Eq. (1.1), a transfer contributes a positive destination-row entry and an equal loss on the source diagonal. This construction connects the individual processes to particle conservation and to the stationary and transient solutions.

For bound levels $i \neq j$, define $q_{j\to i}$ as the corresponding electron-impact coefficient. Then

$$C_{ij} = A_{j\to i} + n_\mathrm{e} q_{j\to i}, \tag{3.1}$$

$$C_{ii} = -\sum_{j\neq i} (A_{i\to j} + n_\mathrm{e} q_{i\to j}) - n_\mathrm{e} \sum_p S_{ip}. \tag{3.2}$$

Radiative entries vanish for upward or forbidden transitions. Ionization and recombination couple the bound levels to adjacent-charge parents through

$$C_{pi} = n_e S_{ip}, \qquad C_{ip} = n_e R_{ip}. \tag{3.3}$$

The parent diagonal contains the corresponding recombination loss into the lower stage. When all destinations belong to the modeled system, the gain and loss terms cancel in each source column:

$$\sum_i C_{ij} = 0 \quad \text{for every source column } j, \qquad \frac{\mathrm{d}}{\mathrm{d}t}\sum_i N_i = 0. \tag{3.4}$$

An open subsystem may have nonzero column sums, provided its external sources and sinks account for the population that crosses the boundary.

Implementation. populate_cr_matrix stores the main generator in `cr_matrix['cr']`; `A_ji` is stored with upper-level index first, so radiative entries are transposed when inserted into the generator. `cr_red` later contains only the selected excited-state block. The separate `cr_loss` array holds only the level-loss diagonal of that generator.

The quasistatic CX columns represent forcing by prescribed parent populations and therefore contain gains without matching losses. For full evolution, solve_time_dependent assigns those gains to the recombining-parent columns and includes the parent depletion on the diagonal. The resulting generator obeys Eq. (3.4). The additional CX bookkeeping columns are then empty and require zero initial populations and zero sources.

Nonnegative off-diagonal entries, the expected column sums and the correct density powers are necessary checks on the kinetic model. They test the physical assembly independently of the ability of a linear solver to return a numerical solution.

3.1 Time-dependent evolution and matrix exponentials

For fixed plasma conditions and an optional constant source s,

$$\dot{N} = CN + s, \qquad N(t) = \mathrm{e}^{Ct}N(0) + \int_0^t \mathrm{e}^{C(t-u)}s\,\mathrm{d}u. \tag{3.5}$$

The exponential propagates the initial populations, while the integral accumulates injection from the source throughout the interval. For absolute populations, s has units of density per unit time. This solution is exact for fixed C and constant s; numerical evaluation is discussed in Appendix B.

All requested times refer to the initial state at $t = 0$. The choice of that state encodes the physical preparation of the species. In the surface-release example [1], an initially ground-state population develops metastable populations and time-dependent emission. Relating this evolution to distance from the surface additionally requires particle velocities and a transport geometry.

3.2 Stationary solutions

A normalized stationary population of a closed model satisfies

$$C\pi = 0, \qquad \mathbf{1}^{\mathsf{T}}\pi = 1, \qquad \pi_i \geq 0. \tag{3.6}$$

For a unique stationary state, normalization replaces one redundant balance equation. Rescaling the balance equations can help preserve constraints associated with weak rates. If several disconnected closed classes are present, normalization alone leaves their relative weights undetermined; those weights depend on the initial condition.

Multiplication of π by the conserved population recovers absolute densities in a source-free closed system. With a prescribed source, stationarity instead requires $CN + s = 0$ and compatible population losses or boundary conditions. A stationary collisional-radiative state may remain far from LTE because it balances radiative escape as well as collisions. The numerical stationary algorithms are described in Appendix B.

3.3 A two-level density scan

For a two-level ion with upward and downward coefficients q_{01}, q_{10} and radiative rate A, the excited population obeys

$$\dot{N}_1 = n_e q_{01} N_0 - (A + n_e q_{10}) N_1, \qquad N_0 + N_1 = N_{\text{ion}}. \tag{3.7}$$

At stationarity, excitation is balanced by radiative and collisional return, giving $N_1/N_0 = n_e q_{01}/(A + n_e q_{10})$. Normalization to the conserved ion population yields

$$f_1 = \frac{n_e q_{01}}{A + n_e(q_{01} + q_{10})}, \qquad \text{PEC}_{10} = \frac{A q_{01}}{A + n_e(q_{01} + q_{10})}. \tag{3.8}$$

The photon emissivity is $AN_1 = n_e N_{\text{ion}} \text{PEC}_{10}$. Thus the PEC in this example is defined per ion total; a coefficient per ground population would instead be $Aq_{01}/(A + n_e q_{10})$. At low density, the total-normalized PEC approaches q_{01} and the excited fraction grows linearly with density. At high density, $f_1 \to q_{01}/(q_{01} + q_{10})$ and the PEC decreases as $1/n_e$. The photon rate per ion nevertheless approaches the finite value Af_1. The decrease of a density-normalized coefficient therefore does not imply a vanishing photon yield.

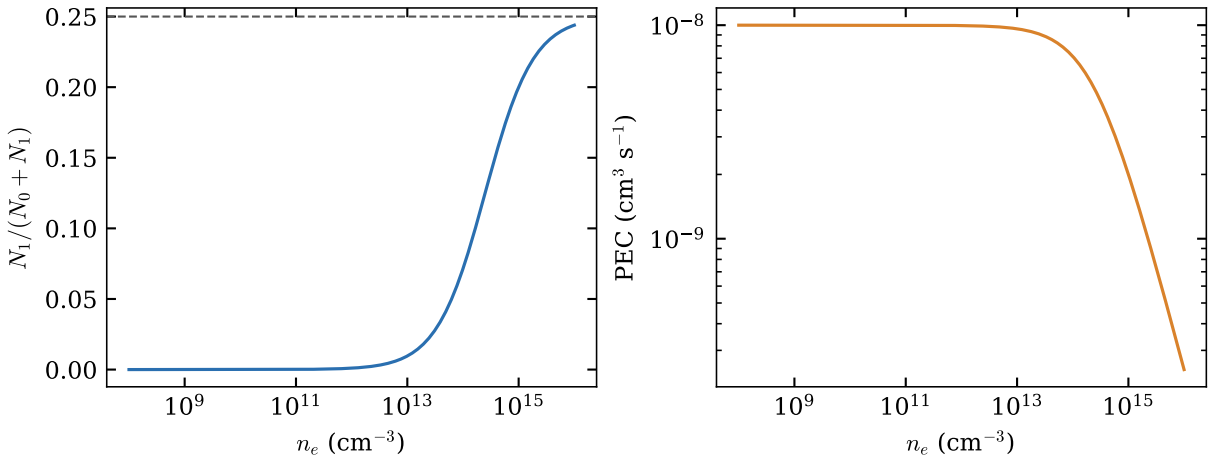


Figure 2: An analytic illustration with $A = 10^7\,\text{s}^{-1}$, $q_{01} = 10^{-8}$ and $q_{10} = 3 \times 10^{-8}$ cm^3/s. The dashed line is the high-density fraction 1/4. These are illustrative coefficients, not a fit to a specific atom or an execution of the ColRadPy atomic-data path.

3.4 A conservative chain and a slow mode

Consider three states connected by irreversible unit-rate transfers, $0 \to 1 \to 2$. With time measured in seconds and rates in s^{-1},

$$C = \begin{pmatrix} -1 & 0 & 0 \\ 1 & -1 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad N(0) = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad N(t) = \begin{pmatrix} \text{e}^{-t} \\ t\text{e}^{-t} \\ 1 - (1+t)\text{e}^{-t} \end{pmatrix}. \tag{3.9}$$

State 1 initially accumulates population and subsequently empties into state 2, while the total population remains constant. Both the ColRadPy solver and an independent SciPy exponential recover the analytic solution. The te^{-t} contribution also tests the treatment of a defective eigenbasis, discussed in Appendix B.

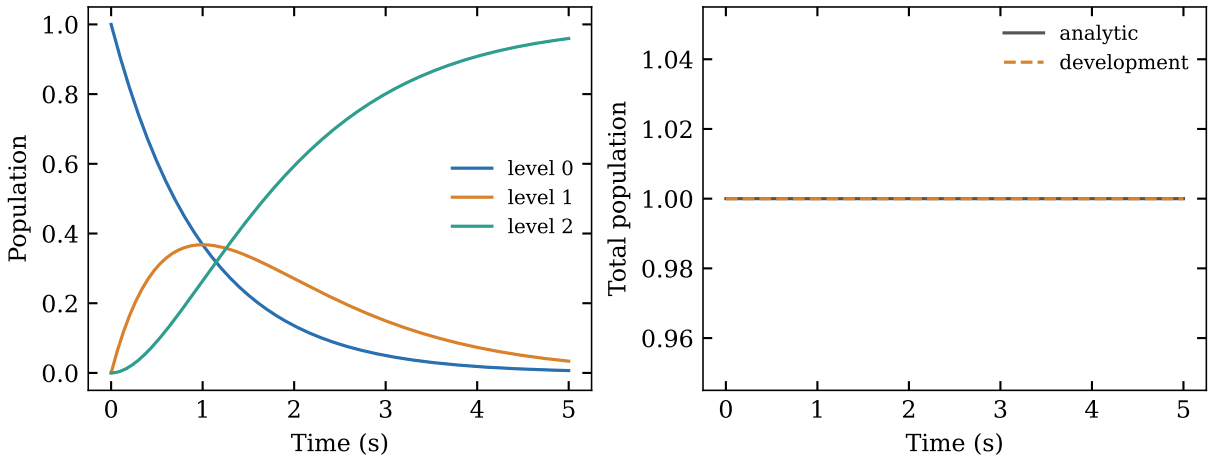


Figure 3: Equal-rate chain: solid curves are analytic populations; dashed curves are ColRadPy results. The right panel tests the conserved sum. Agreement includes the te^{-t} contribution from the defective basis.

A complementary check concerns a very slow transition. For $C = [[-r, 0], [r, 0]]$ with $r = 10^{-12}$ s^{-1} and $N(0) = (1, 0)$, the remaining initial-state population at 10^{12} s is e^{-1}. The source solver with zero source reproduces this evolution and agrees with the source-free calculation, confirming that the slow mode is retained over the relevant observation time.

4 Quasistatic reduction and population normalization

The full kinetic system often contains excited levels that relax much faster than the retained ground, metastable and parent populations. Quasistatic elimination exploits this separation by replacing the fast evolution equations with instantaneous balance. The excited populations then follow the retained populations through a linear response, and the remaining dynamics can be expressed with effective rates.

Partition the population vector into retained states N_r and eliminated excited states N_e. For a closed reduction, the retained set includes both the selected bound levels and the adjacent-charge parents. In this partition the full equations take the block form

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{pmatrix} N_r \\ N_e \end{pmatrix} = \begin{pmatrix} C_{rr} & C_{re} \\ C_{er} & C_{ee} \end{pmatrix}\begin{pmatrix} N_r \\ N_e \end{pmatrix}. \tag{4.1}$$

When excited-state relaxation is fast compared with evolution of the drivers,

$$0 = C_{ee}N_e + C_{er}N_r, \qquad N_e = -C_{ee}^{-1}C_{er}N_r. \tag{4.2}$$

The matrix $-C_{ee}^{-1}C_{er}$ specifies the response to each retained population. Its existence requires an invertible excited block, while its physical use requires that the block relax on a sufficiently short timescale. A slowly decaying eliminated mode can invalidate the approximation even when the inverse is well defined. The relevant time-scale and emission-convergence criteria are developed below.

Substitution into the retained equations gives the Schur complement

$$C_{\text{eff}} = C_{rr} - C_{re}C_{ee}^{-1}C_{er}. \tag{4.3}$$

The retained block contains the direct processes and losses, while the second term accounts for excursions through the eliminated excited manifold. Together they define the effective coefficients of Section 5. Column-sum conservation is preserved for a closed model, although the sum of the retained variables must still be distinguished from a total that includes reconstructed excited populations (Section 4.4). The inverse notation is convenient for the derivation; numerically the same response can be obtained by solving $C_{ee}Y = -C_{er}$.

4.1 Choosing retained levels and testing time-scale separation

The GCR hierarchy distinguishes rapidly decaying autoionizing states, ordinary excited levels and more slowly evolving ground and metastable populations [9]. At finite density, collisional redistribution competes with radiative decay and autoionization, so the relevant relaxation times belong to the coupled kinetic system. The reduction is appropriate where these times are short compared with the timescales on which the retained populations and plasma conditions change. High-shell projection applies the same reasoning to an auxiliary excited manifold (Section 7).

Spectroscopic metastability is consequently an initial guide to the retained set, rather than a sufficient dynamical criterion [1]. A weak radiative decay can permit substantial population accumulation, while collisions and ionization shorten the effective relaxation time. Conversely, strongly coupled ordinary excited levels may support a slow collective mode. The retained manifold therefore depends on the plasma conditions, initial state and observable under investigation.

For constant matrix blocks, write the instantaneous quasistatic response as $N_e^{\text{qs}}(t) = -C_{ee}^{-1}C_{er}N_r(t)$ and let $\delta = N_e - N_e^{\text{qs}}$. The exact lag equation is

$$\dot{\delta} = C_{ee}\delta - \dot{N}_e^{\text{qs}}, \tag{4.4}$$

$$\delta(t) = \mathrm{e}^{C_{ee}t}\delta(0) - \int_0^t \mathrm{e}^{C_{ee}(t-u)}\dot{N}_e^{\text{qs}}(u)\,\mathrm{d}u. \tag{4.5}$$

The first term describes relaxation of the initial excited-state departure from quasistatic balance; the integral describes the lag induced by evolving drivers. After the initial transient has decayed, accuracy depends on the ratio of excited relaxation time to driver evolution time. Eigenvalues and eigenvectors of C_{ee} characterize this relaxation, subject to possible transient amplification in a nonnormal system. A spectral gap alone is therefore insufficient as an error bound. When plasma conditions vary, the response matrix also changes and contributes to $\dot{N}_e^{\text{qs}}$.

The adequacy of a retained manifold can be assessed by comparing full and reduced solutions initialized with consistent retained populations, then enlarging the manifold until the predicted emission converges over the observation interval. Population magnitude alone is insufficient: a weakly populated level may be an important feeder of a diagnostic line. Relaxation modes are combinations of levels, so associating a mode's decay rate with a single level requires inspection of its eigenvector.

Published comparisons for Be I and Mo I illustrate this dependence on species and conditions [1]. Table 2 summarizes the effect of retaining metastables on the time-dependent line emission relative to full level evolution.

The clearer separation between slow and fast modes in Be permits an effective reduction with a small retained manifold. The more gradual relaxation spectrum in Mo makes that reduction less successful. These examples motivate convergence tests for the chosen observable and conditions, rather than a fixed number of retained levels for either species. A ground-only reduction also removes the independent evolution of metastable fractions.

Published case	Plasma conditions	Reported observation
Be I	$T_e = 20$ eV, $n_e = 10^{13}$ cm^{-3}	Retaining the important metastable substantially improves the time-dependent line-emission prediction over a ground-only reduction.
Mo I	$T_e = 8$ eV, $n_e = 5 \times 10^{11}$ cm^{-3}	Even a ten-metastable reduction does not reproduce the full time-dependent line history.

Table 2: Retained sets compared with the full time-dependent model [1], not rerun for this manual. They do not set a universal number of retained levels for either species.

4.2 Separate excitation, recombination and CX drivers

The retained populations drive excited levels through distinct physical channels. Separating bound-level forcing, electron recombination and charge exchange gives

$$0 = C_{ee}N_e + C_{em}N_m + n_e R_{ep} N_p + n_h K_{ep} N_p^{CX}. \tag{4.6}$$

The corresponding dimensionless responses are

$$F = -C_{ee}^{-1} C_{em}, \qquad G = -C_{ee}^{-1} n_e R_{ep}, \qquad H = -C_{ee}^{-1} n_h K_{ep}, \tag{4.7}$$
$$N_e = F N_m + G N_p + H N_p^{CX}.$$

Each column describes the population response of the eliminated manifold to one prescribed driver. Multiplication by the actual driver populations and summation recover the physical excited populations. A response column has no probability normalization of its own.

Implementation. solve_quasi_static stores the forcing as `beta`$= -C_{er}$ and evaluates `pops`$= C_{ee}^{-1}$`beta`. Columns are ordered as excitation, recombination and CX when enabled. `processed['F']` omits the additional inverse-density factor of some ADAS conventions and follows the selected excitation normalization. It equals the unnormalized F in Eq. (4.7) when normalization is disabled; the mapping to other stored conventions is given in Section 4.3.

CX-only construction. When CX is enabled but both radiative and three-body recombination are disabled, the CX columns still follow the parent columns, which the generator keeps whether or not recombination is used.

4.3 Normalization is a choice of driving population

Normalization specifies the population to which a response or emission coefficient refers. Absolute level densities are recovered by weighting the independent response columns in Eq. (4.7); level fractions additionally require a defined population total. The conventions most frequently used below are summarized in the table.

For a single ground driver and no capture contribution, $N_i = F_{i0} N_0$ and $N_{ion} = N_0 + \sum_i N_i$. Therefore

$$\frac{N_0}{N_{ion}} = \frac{1}{1 + \sum_i F_i}, \qquad \frac{N_i}{N_{ion}} = \frac{F_i}{1 + \sum_j F_j}. \tag{4.8}$$

Here $F_i = F_{i0}$. The rescaling converts an excitation response per ground population into a level fraction per ion total. Recombination responses retain a parent population from the next charge stage as their denominator. For example, parent-driven injection $n_e \alpha = 2$ s^{-1} into a level with loss rate $A = 1$ s^{-1} gives $N_i/N_p = 2$. Such a value is consistent with a linear response and illustrates the importance of preserving the separate GCR drivers [2].

Quantity	Denominator	Interpretation
N_i	None	Absolute level density, cm^{-3}
$F_{i\rho}$	Retained bound driver N_ρ	Excited response of that driver's column
$G_{ip},\ H_{ip}$	Parent driver N_p or N_p^{CX}	Capture-driven response in another charge stage
N_i/N_{stage}	Total population in the stage	Level fraction after all contributions are assembled

Table 3: Population conventions. Responses are dimensionless but their column sums need not be one.

For an excitation driver ρ, define

$$d_\rho = \begin{cases} (1+\sum_{i\in e} F_{i\rho})^{-1}, & \text{excitation normalization on,} \\ 1, & \text{excitation normalization off.} \end{cases} \tag{4.9}$$

In this definition F is the unnormalized response of Eq. (4.7). The normalization applied to stored excitation quantities is consequently

$$\begin{aligned} F_{i\rho}^{\mathrm{stored}} &= d_\rho F_{i\rho}, & \mathrm{PEC}_{u\ell,\rho}^{\mathrm{stored,exc}} &= d_\rho \mathrm{PEC}_{u\ell,\rho}^{\mathrm{exc}}, \\ \mathcal{S}_{\rho p}^{\mathrm{stored}} &= d_\rho S_{\rho p} + \sum_i S_{ip} F_{i\rho}^{\mathrm{stored}} = d_\rho \mathcal{S}_{\rho p}. \end{aligned} \tag{4.10}$$

Both the direct and excited-mediated ionization terms carry the same factor d_ρ. Using a normalized `processed['F']` with an unscaled direct term would therefore mix population conventions. The original response columns are available in `processed['pops_no_norm']`, with excitation columns first. Recombination and CX responses G, H are unaffected by this excitation normalization.

For a single ground driver without capture, $d_0 = N_0/N_{\mathrm{ion}}$, and the normalized excitation coefficient multiplies N_{ion}. With several retained levels, separate normalization of each column describes its driver and associated excited response in isolation. A physical mixture is instead assembled from unnormalized responses and the actual retained populations.

Selecting the stored convention. Excitation normalization defaults on for one retained metastable and off for several; parent-driven recombination and CX columns keep their independent denominators. `default_pop_norm=False` reverses this selection. The actual convention is recorded in `processed['driving_populations_norm']`. Effective-coefficient derivations use unnormalized responses, with the stored SCD mapping specified by Eq. (4.10).

4.4 Retained fractions and total ion fractions

Normalization of retained populations across all modeled charge stages gives $f_a^{(r)} = N_a/\sum_{b\in r} N_b$. Converting these retained fractions to fractions of the full modeled population requires the excited-state contribution. For one charge stage the reconstructed bound-level total is

$$N_{\mathrm{stage}} = \sum_\rho N_\rho + \sum_{i\in e}\left(\sum_\rho F_{i\rho}N_\rho + \sum_p G_{ip}N_p + \sum_p H_{ip}N_p^{\mathrm{CX}}\right). \tag{4.11}$$

All terms in this reconstruction use unnormalized responses and a common absolute scale for the driving populations. The sum includes the explicit bound levels; populations in the projected high shells remain eliminated and are not counted separately.

When the omitted excited population is small, retained and total fractions are close. Otherwise, total ion fractions and power per ion of the element require reconstruction of each stage and normalization by the sum of those stage totals. Excitation-normalized single-ground coefficients account for the excited population associated with that driver, while capture-driven excited populations remain parent-driven responses (Section 5.7). Equation (4.11) makes both contributions explicit.

This distinction also affects transients. The Schur complement conserves the retained sum, whereas the reconstructed total may vary because the eliminated population derivatives were neglected in the reduced equations. Appreciable transient storage in excited levels therefore calls for a larger retained manifold or full level evolution; renormalization alone does not supply the missing dynamics.

5 Effective coefficients and charge-state evolution

The Schur complement converts the response of the eliminated levels into effective transfers between retained states. Its bound-to-parent, parent-to-bound and within-manifold blocks define the GCR coefficients summarized below. Division by the appropriate bath density expresses these rates in cm^3/s. Throughout the derivation, the responses of Eq. (4.7) are unnormalized and the retained populations are prescribed explicitly. Stored array orientations are documented in Appendix A.1.

Coefficient	Source	Destination	Included pathways
SCD	Bound ρ	Parent p	Direct ionization and ionization after bound excitation
ACD	Parent p	Bound ρ	Electron capture and stabilization through excited levels
CCD	CX parent p	Bound ρ	Donor-driven capture and subsequent stabilization
QCD	Bound σ	Bound ρ	Direct mixing and redistribution through excited levels
XCD	Parent p	Parent q	Capture and re-ionization through the lower ion

Table 4: Physical meaning of the effective transfers. SCD, ACD, QCD and XCD multiply n_e; CCD multiplies n_h. Projection can add the high-shell pathways described in Section 7.

5.1 SCD: effective ionization

For a retained bound level ρ and destination parent p,

$$\mathcal{S}_{\rho p} = S_{\rho p} + \sum_i S_{ip} F_{i\rho}. \tag{5.1}$$

The first term is direct ionization of the retained level; the second accounts for population transferred through excited levels before ionization. The effective contribution to C_eff from ρ to p is $n_\mathrm{e}\mathcal{S}_{\rho p}$.

5.2 ACD and CCD: effective capture

The capture coefficient from parent p into retained level ρ is

$$\mathcal{A}_{\rho p} = R_{\rho p} - \sum_{ij} C_{\rho i} (C_{ee}^{-1})_{ij} R_{jp}. \tag{5.2}$$

The indirect term describes capture into the excited manifold followed by redistribution and radiative stabilization into ρ. Replacing R by K gives the charge-exchange coefficient $\mathcal{C}_{\rho p}$ (CCD).

ACD multiplies electron density and CCD multiplies donor density; an ionization matrix expressed per electron density therefore contains $(n_\mathrm{h}/n_\mathrm{e})\mathcal{C}$ for charge exchange.

5.3 QCD: redistribution within a retained bound manifold

For two different retained bound levels,

$$\mathcal{Q}_{\sigma\to\rho} = \frac{C_{\rho\sigma} + \sum_i C_{\rho i} F_{i\sigma}}{n_\mathrm{e}}, \qquad \rho \neq \sigma. \tag{5.3}$$

QCD includes both direct retained-level mixing and redistribution through the excited manifold. Its stored diagonal is zero; the corresponding source losses are included when the charge-state generator is assembled.

5.4 XCD: redistribution between parent metastables

A parent p can recombine into an excited level of the lower stage and then ionize into a different parent q. The coefficient for this path is

$$\mathcal{X}_{p\to q} = -\sum_{ij} C_{qi}(C_{ee}^{-1})_{ij} R_{jp} = -n_\mathrm{e} \sum_{ij} S_{iq}(C_{ee}^{-1})_{ij} R_{jp}, \qquad p \neq q. \tag{5.4}$$

The factor n_e arises because ionization out of the excited manifold is a density-weighted rate. The resulting units are

$$[n_\mathrm{e} S][C_{ee}^{-1}][R] = (\mathrm{s}^{-1})(\mathrm{s})(\mathrm{cm}^3\,\mathrm{s}^{-1}) = \mathrm{cm}^3\,\mathrm{s}^{-1}. \tag{5.5}$$

Equivalently, XCD is the off-diagonal parent block of the Schur complement divided by electron density. Its contraction requires a common eliminated-level ordering for the recombination rows and C_{ee}. The density factor agrees with Eq. 17 of Summers et al. [9]. High-shell projection adds the parent transfer X_{qp}^{ind} defined in Eq. (7.6). This correction is stored with destination first and is transposed on addition to the source-first XCD array (Appendix A.1).

XCD contraction. The contraction uses $n_\mathrm{e}S$ and selects recombination rows by the actual eliminated-level map, including noncontiguous metastable selections such as $[0, 2]$. The resulting coefficient has units cm^3/s and agrees with the parent block of the direct Schur complement divided by electron density.

5.5 XCD checked by direct elimination

The bundled Be II file is solved with metastables $[0, 1]$, temperatures $[5, 20]$ eV and densities $[10^{12}, 10^{14}]$ cm^{-3}, with both recombination mechanisms and ionization enabled. At the first grid point, the off-diagonal parent block of Eq. (4.3), formed with a linear solve rather than the stored inverse and divided by n_e, agrees with the stored XCD$[1, 0]$ (Table 7); both are in cm^3/s. A separate test with the noncontiguous metastables $[0, 2]$ covers the row selection.

5.6 Ionization balance across charge stages

Effective coefficients from successive ions form a coupled charge-state model. Collecting their retained populations into x gives, at fixed plasma conditions,

$$\dot{x} = Mx + s, \qquad M = n_\mathrm{e} B. \tag{5.6}$$

The coefficient matrix B has units cm^3/s. SCD and ACD couple adjacent stages, while QCD redistributes retained populations within the ion from which it was calculated. XCD adds redistribution within the parent stage through excursions into the lower ion. A given stage can therefore receive its own QCD contribution and the XCD contribution calculated from the preceding stage.

5.6.1 Matrix assembly and density factors

Each off-diagonal coefficient is accompanied by an equal source-diagonal loss, giving $\mathbf{1}^{\mathsf{T}}B = 0$ for a closed model. Charge-exchange entries contain $(n_{\mathrm{h}}/n_{\mathrm{e}})\mathcal{C}$, so multiplication of B by n_{e} restores the donor-density rate $n_{\mathrm{h}}\mathcal{C}$.

Implementation. populate_ion_matrix stores B in `ion_matrix`, despite that name suggesting a density-weighted rate. solve_no_source and solve_source multiply by the electron-density grid before invoking the matrix-exponential solvers. Do not apply the density multiplier a second time when interpreting their populations.

5.6.2 Ionization age

For constant electron density, the ionization age is $\tau = n_{\mathrm{e}}t$. At fixed temperature, density-independent B gives $x(\tau) = \exp(B\tau)x(0)$. In a GCR model, collisional quenching and three-body capture can introduce an explicit density dependence into B. Equal ionization age then need not produce equal charge-state fractions at different densities.

Ionization-age grids. The wrapper constructs times as $t = \tau/n_{\mathrm{e}}$ with axes *density, time* when there are multiple densities; one density uses a one-dimensional time array. This agrees with the matrix-exponential solver's grid convention, including singleton axes.

5.7 Stage-to-stage coefficients

Charge-state kinetics can retain separate metastable populations or reduce each stage to one population. Two reductions are ground condensation and contraction with prescribed metastable fractions. Each requires a prescription for the populations within the stage, and their treatment of excitation-fed and capture-fed populations determines the meaning of the resulting stage coefficients [9].

5.7.1 Resolved metastables and ground condensation

In the resolved description, each selected ground or metastable level has an independent population. QCD and XCD determine redistribution, and SCD, ACD and CCD connect adjacent stages. The within-stage fractions consequently follow from the coupled kinetics.

Ground condensation eliminates all bound levels except the ground of each stage. Under the single-driver normalization of Section 4.3, the stage variable includes the excited population driven by ground excitation. Capture-driven excited populations remain responses to the next stage, and ACD measures the capture flow reaching the retained ground. This conventional CR reduction therefore uses the stage-population convention described in Section 4.4.

5.7.2 Contraction with metastable fractions

A second construction contracts resolved coefficients with prescribed metastable fractions. Writing the ground-relative ratios as $\eta_\rho = N_\rho/N_g$ with $\eta_g = 1$ gives $f_\rho = \eta_\rho / \sum_\sigma \eta_\sigma$. If f_p^+ denotes the corresponding fractions in the recombining stage, the contracted coefficients are

$$\mathcal{S}_{\text{stage}} = \sum_\rho f_\rho \sum_p \mathcal{S}_{\rho p}, \qquad \text{PLT}_{\text{stage}} = \sum_\rho f_\rho \, \text{PLT}_\rho, \qquad \mathcal{A}_{\text{stage}} = \sum_p f_p^+ \sum_\rho \mathcal{A}_{\rho p}. \tag{5.7}$$

SCD and PLT are weighted by the fractions of the ionizing or emitting stage. ACD is weighted by the recombining-parent fractions and summed over bound destinations; CCD uses the same construction. The parent fractions also weight PRB and PRC. The resulting model has one population per stage, with no separate QCD or XCD entries, although redistribution can influence the fractions used in the contraction.

The ADAS GCR-project contraction uses MET ratios obtained from a within-stage model with a prescribed ground driver and no capture feeding from the next stage (Appendix C). For one metastable μ, its quasistatic balance gives

$$\eta_\mu = \frac{\mathcal{Q}_{g\to\mu}}{\mathcal{Q}_{\mu\to g} + \sum_p \mathcal{S}_{\mu p}}. \tag{5.8}$$

With several metastables, their responses to the ground driver are obtained from a linear system. These excitation-only ratios describe the specified within-stage balance and can differ from the fractions established by capture and ionization in a coupled plasma.

5.7.3 When capture feeding changes the fractions

The effect of capture feeding can be seen directly in the same ground–metastable system. Including capture from the next stage, while neglecting CX and other feeding channels in this illustration, gives the stationary metastable balance

$$0 = \mathcal{Q}_{g\to\mu} N_g - \left(\mathcal{Q}_{\mu\to g} + \sum_p \mathcal{S}_{\mu p} \right) N_\mu + \sum_p \mathcal{A}_{\mu p} N_p^+, \tag{5.9}$$

after dividing the rate equation by n_e. It follows that

$$\frac{N_\mu}{N_g} = \eta_\mu + \frac{\sum_p \mathcal{A}_{\mu p} N_p^+}{(\mathcal{Q}_{\mu\to g} + \sum_p \mathcal{S}_{\mu p}) N_g}. \tag{5.10}$$

The additional term is negligible when capture feeding is small relative to $\mathcal{Q}_{g\to\mu} N_g$. The parent abundance alone is therefore insufficient to assess the excitation-only approximation. Substantial capture changes the metastable population available for re-ionization, whereas ground condensation assigns that population through a different retained variable. The accuracy of either reduction depends on the ion and conditions. In the resolved system, XCD from the preceding stage also contributes to redistribution of these populations [9].

Stage contraction. For ADF04 inputs, `use_met=True` in ionization_balance contracts resolved coefficients using fractions from metastable_fractions, calculated from each stage's QCD and SCD. The default `use_met=False` preserves the supplied retained manifolds. Matching unresolved ADAS coefficients additionally requires consistent MET data, interpolation and atomic inputs.

Retaining the metastables allows these fractions to evolve within the coupled model. Its accuracy remains subject to the atomic data, time-scale separation and population conventions specified above.

6 Emission, power and spectroscopic diagnostics

Spectroscopic observables follow directly from the level populations. A one-photon transition produces photons at rate $A_{u\ell}N_u$ per unit volume, and each photon carries energy $\Delta E_{u\ell}$. Substitution of the quasistatic responses separates the emissivity into excitation, recombination and charge-exchange contributions. Their coefficients inherit the driving-population convention of Section 4.3.

To compare notation, the response functions used in Section 1.2.2 of Summers et al. [9] are $\mathcal{F}^{(\mathrm{exc})}_{u\rho} = F_{u\rho}/n_\mathrm{e}$ and $\mathcal{F}^{(\mathrm{rec})}_{up} = G_{up}/n_\mathrm{e}$. They include an additional inverse electron-density factor relative to the dimensionless responses defined here. Stored excitation quantities may also carry the normalization of Eq. (4.9).

6.1 Photon emissivity coefficients

For optically thin emission, the angle-integrated photon emissivity of a one-photon line $u \to \ell$ is $A_{u\ell}N_u$. Resolving the upper-level population by driver gives

$$\epsilon_{u\ell} = A_{u\ell}N_u = n_\mathrm{e}\sum_\rho N_\rho \mathrm{PEC}^{\mathrm{exc}}_{u\ell,\rho} + n_\mathrm{e}\sum_p N_p \mathrm{PEC}^{\mathrm{rec}}_{u\ell,p} + n_\mathrm{h}\sum_p N^{\mathrm{CX}}_p \mathrm{PEC}^{\mathrm{CX}}_{u\ell,p} \quad [\mathrm{photons\,cm^{-3}\,s^{-1}}]. \tag{6.1}$$

Factoring out the bath density and driving population defines the photon emissivity coefficients, in cm^3/s:

$$\mathrm{PEC}^{\mathrm{exc}}_{u\ell,\rho} = A_{u\ell}F_{u\rho}/n_\mathrm{e}, \tag{6.2}$$

$$\mathrm{PEC}^{\mathrm{rec}}_{u\ell,p} = A_{u\ell}G_{up}/n_\mathrm{e}, \tag{6.3}$$

$$\mathrm{PEC}^{\mathrm{CX}}_{u\ell,p} = A_{u\ell}H_{up}/n_\mathrm{h}. \tag{6.4}$$

For a retained upper level σ, the population is already an independent variable. Its response to a retained bound driver ρ is $\delta_{\sigma\rho}$, so its radiative transitions have

$$\mathrm{PEC}^{\mathrm{exc}}_{\sigma\ell,\rho} = \delta_{\sigma\rho}A_{\sigma\ell}/n_\mathrm{e}, \qquad \mathrm{PEC}^{\mathrm{rec}}_{\sigma\ell,p} = \mathrm{PEC}^{\mathrm{CX}}_{\sigma\ell,p} = 0. \tag{6.5}$$

The same emissivity can be expressed per ion total by changing the coefficient's denominator. For a single ground driver without capture, the corresponding excitation PEC is $d_0\mathrm{PEC}^{\mathrm{exc}}$, where $d_0 = N_0/N_\mathrm{ion}$. For a prescribed mixture it is $\sum_\rho (N_\rho/N_\mathrm{ion})\mathrm{PEC}^{\mathrm{exc}}_{u\ell,\rho}$. The factor is applied once, consistently with the population that multiplies the coefficient. Capture and CX retain their own parent drivers. For an isotropic emitter, the line-of-sight photon radiance is $\int \epsilon_{u\ell}\,\mathrm{d}s/(4\pi)$; angular dilution is therefore introduced at the observation stage. Section 6.4 relates the excitation component to ionizations per emitted photon.

CX PEC convention. The implementation divides *all* computed driving columns by n_e when forming `processed['pecs']`. Its stored CX columns therefore contain $AH/n_\mathrm{e} = (n_\mathrm{h}/n_\mathrm{e})\mathrm{PEC}^{\mathrm{CX}}$ as defined above. Use electron density with that stored response, or explicitly convert to a donor-normalized coefficient before using Eq. (6.1).

6.2 Two-photon decay and the limits of line outputs

Two-photon decay requires a spectral distribution as well as a total transition rate. A decay rate $A^{2\gamma}_{u\ell}$ removes population at $A^{2\gamma}_{u\ell}N_u$, produces photons at twice that rate and radiates total power $\Delta E_{u\ell}A^{2\gamma}_{u\ell}N_u$. The two photon energies sum to the level separation. If $p(\nu)$ is the normalized

distribution of individual photon frequencies, $\int p\,\mathrm{d}\nu = 1$ and $\int h\nu p\,\mathrm{d}\nu = \Delta E_{u\ell}/2$, the spectral emissivity is

$$\epsilon_\nu^{2\gamma} = 2A_{u\ell}^{2\gamma} N_u\, p(\nu). \tag{6.6}$$

The distribution p contains atomic information beyond the integrated decay rate [14]. A total A-value therefore determines the population loss and integrated power but not the continuum shape.

FAC two-photon output. FAC import includes the two-photon contribution in the pair's A-value and records `two_photon` metadata. Generic PEC, wavelength, profile and ADF15 outputs treat that entry as a one-photon transition. For its two-photon component, AN_u therefore counts decay events: the assigned wavelength $hc/\Delta E$ is not a continuum spectrum and is unsuitable for a line S/XB diagnostic. Energy weighting still recovers the integrated power $\Delta E A N_u$. A photon spectrum requires separation of this contribution using the metadata, an appropriate continuum distribution and a separate export.

6.3 Radiated power coefficients

Multiplying a PEC by the photon energy $\Delta E_{u\ell} = hc/\lambda_{\mathrm{vac}}$ in joules gives a line-power coefficient in W cm^3; conversion to W m^3 introduces a factor 10^{-6}. Summation over transitions yields the excitation and recombination power coefficients (Summers et al., Eqs. 24–25):

$$\mathrm{PLT}_\rho = \sum_{u\ell} \Delta E_{u\ell}\,\mathrm{PEC}^{\mathrm{exc}}_{u\ell,\rho} = \sum_{u\in e}\sum_{\ell<u} \frac{\Delta E_{u\ell} A_{u\ell} F_{u\rho}}{n_{\mathrm{e}}} + \sum_{\ell<\rho} \frac{\Delta E_{\rho\ell} A_{\rho\ell}}{n_{\mathrm{e}}}, \tag{6.7}$$

$$\mathrm{PRB}_p = \sum_{u\ell} \Delta E_{u\ell}\,\mathrm{PEC}^{\mathrm{rec}}_{u\ell,p} = \sum_{u\in e}\sum_{\ell<u} \frac{\Delta E_{u\ell} A_{u\ell} G_{up}}{n_{\mathrm{e}}}. \tag{6.8}$$

Here $u \in e$ denotes the eliminated upper levels. The CX power coefficient PRC_p is obtained analogously from $\mathrm{PEC}^{\mathrm{CX}}$. The final term in PLT represents radiation from the retained driver ρ itself, as given by Eq. (6.5). Summers et al. treat it separately and describe it as usually small in fusion plasmas; ADAS includes it in PLT. Its magnitude depends on the retained level and can be substantial.

For example, a calculation for the Ne^0 2p^{5}3s ^{3}P term uses retained levels $[0, 1]$ and three OPEN-ADAS files, which are not distributed with ColRadPy: the ADF04 file `cop98#10_ls#ne0.dat`, the projection mapping `exp96#ne_ne0ls.dat` and its ADF17 file `cbnm96#ne_ne0ls.dat`. At $T_{\mathrm{e}} = 10$ eV, radiation with ^{3}P as the upper term accounts for 98 % of that driver's PLT at $n_{\mathrm{e}} = 10^{12}$ cm^{-3} and 44 % at 10^{14} cm^{-3}. These fractions are the sum of `pls` over the relevant lines divided by `plt` for that driver.

The definition of recombination power also depends on the included radiation channels. Summers et al. include bremsstrahlung in PRB, and ADAS additionally includes the recombination continuum and high-shell radiation (Appendix C). ColRadPy's `prb` is restricted to Eq. (6.8): radiation from the explicit ADF04 levels following capture.

Power outputs. `processed['pls']` contains the energy-weighted PEC for each line and driver. Summation over lines, separately for each excitation, recombination or CX driver, gives `plt`, `prb` and `prc` in W cm^3. Photon energies follow from level-energy differences. Level pairs are ordered by lower then upper level in `pec_levels`, including retained upper levels; zero radiative rates contribute zero emission. The time-dependent `['td']['pecs']`, `pls` and `plt` use the same transitions with the evolving level populations.

6.4 S/XB: ionizations per emitted photon

The diagnostic ratio S/XB relates ionization to excitation-driven line emission [1]. For a single retained driver ρ, define the total effective ionization coefficient $\mathcal{S}_\rho = \sum_p \mathcal{S}_{\rho p}$. For line $u \to \ell$,

$$(S/XB)_{u\ell,\rho} = \frac{\mathcal{S}_\rho}{\mathrm{PEC}^{\mathrm{exc}}_{u\ell,\rho}}. \tag{6.9}$$

The numerator sums the driver's SCD over destination parents, as in Eq. 23 of Summers et al. [9]. With the same density and population convention in numerator and denominator, the ratio is dimensionless and measures ionizations per excitation photon. A parent-restricted numerator defines the corresponding restricted ratio and should be identified explicitly.

For several retained populations, the observable rate ratio becomes

$$(S/XB)_{u\ell}[N_m] = \frac{\sum_\rho N_\rho \mathcal{S}_\rho}{\sum_\rho N_\rho \mathrm{PEC}^{\mathrm{exc}}_{u\ell,\rho}}. \tag{6.10}$$

The weights involve the actual retained populations and each driver's emissivity. Consequently, different metastable mixtures can give different ionization-to-photon ratios at identical temperature and density.

When excitation dominates the observed line and S/XB is approximately uniform along the emitting column, its photon radiance $I_{u\ell}$ determines the integrated ionization rate through

$$\int \dot{n}_{\mathrm{ion}}\, \mathrm{d}s = (S/XB)_{u\ell}\, 4\pi I_{u\ell}. \tag{6.11}$$

For a spatially varying ratio, the ionization rate is obtained by integrating the local product of S/XB and emissivity. Conversion of this column rate into surface erosion or particle influx additionally depends on transport and geometry. Escape before ionization, repeated recombination/ionization and capture-driven photons can all change the connection between entering particles and emitted photons. The ratio is undefined where the excitation PEC vanishes.

S/XB output. S/XB is a ratio of stored results and is not stored itself: `get_sxb` returns Eq. (6.9) for every line and retained driver, from the parent sum of `scd` and the matching excitation column of `pecs`, with the same normalization, and NaN where a line has no excitation emission. `write_sxb_adf13` writes it to an ADF13 file [10], as ADAS208 does, an undefined ratio as the format's floor 10^{-74} with a warning; the file needs a temperature–density grid. Both need ionization in the calculation. A per-driver photon rate in s^{-1} must not be mixed with an ionization coefficient in cm^3/s.

6.5 Unsummed contributions and dominant feeding mechanisms

The response can be decomposed further to identify the channels that feed a particular level [1]. Let a identify an excitation, recombination or CX driver, and let $\beta_{ja} = -C_{ja}$ describe its injection into eliminated level j. Retaining the injection index in the matrix product gives

$$U_{ija} = (C_{ee}^{-1})_{ij}\beta_{ja} \quad (\text{no sum over } j), \qquad N_i = \sum_a \sum_j U_{ija} N_a. \tag{6.12}$$

Summation over j recovers the response to driver a. Individual terms assign its contribution to the level at which population enters the eliminated manifold. For a driver with nonzero response, the relative contributions are

$$f_{j|i,a} = \frac{U_{ija}}{\sum_k U_{ika}} \tag{6.13}$$

and identify the dominant injection channels. Their interpretation requires nonnegative physical rates and a relaxing excited block. A common normalization of a single driver cancels in these fractions, whereas combination of several drivers requires their population weights.

The stored contribution tensor. The implementation stores U in `processed['pop_lvl']` with axes *destination excited level, injection excited level, driver, temperature, density*. Both excited-level indices refer to the reduced level map, not to physical bound-level numbers. Summing the second axis reconstructs `processed['pops']` under the same normalization. Paired mode replaces the last two axes by one pair index.

The diagonal term $j = i$ represents injection directly into the observed upper level, followed by its complete response through C_{ee}^{-1}. Terms with $j \neq i$ describe injection elsewhere and subsequent redistribution or cascading. Since the inverse includes repeated transfers, these are algebraic injection-channel contributions rather than exclusive microscopic paths. In the Mo I examples near 379.8 and 550.6 nm, the decomposition distinguishes direct ground injection from feeding through other populated levels [1]. Comparison with measured intensities requires the population weighting in Eq. (6.1).

The same accounting can resolve effective rates. For example, level i contributes $S_{ip}F_{i\rho}$ to indirect ionization, and contributes $C_{\rho i}G_{ip}/n_e$ to capture feeding retained level ρ. Keeping these terms unsummed can guide a reduced atomic model, provided its emission and effective-rate errors are checked afterward. Bundling levels is another way to reduce the dynamical size of heavy-species models [1]; choosing a short list of retained metastables is not by itself a population-preserving bundling, and ColRadPy provides no automatic bundled model.

6.6 Wavelengths, fine structure and line profiles

6.6.1 Wavelengths and NIST energy matching

Level-energy differences determine vacuum wavelengths through Eq. (1.2). Conversion to an air convention uses $\lambda_{air} = \lambda_{vac}/n(\lambda_{vac})$, where the refractive index is evaluated with the pressure- and temperature-dependent dispersion expression

$$D(\xi) = 8342.54 + \frac{2406147}{130 - \xi^2} + \frac{15998}{38.9 - \xi^2}, \quad \xi = 1000/\lambda_{vac,nm}, \tag{6.14}$$

$$n = 1 + 10^{-8}\left[D(\xi)\frac{P}{96095.43}\frac{1 + 10^{-8}(0.601 - 0.00972T_C)P}{1 + 0.0036610T_C} - P_w(0.037345 - 0.000401\xi^2)\right]. \tag{6.15}$$

Pressure P and water-vapor pressure P_w are in Pa, and T_C is in degrees Celsius. The defaults are 101325 Pa, zero water-vapor pressure and 15 °C. The empirical dispersion formula has poles near 87.706 and 160.334 nm and is unsuitable for unrestricted UV extrapolation. ColRadPy accordingly retains vacuum wavelengths below 200 nm and applies the air conversion above this threshold.

Matched NIST energies refine the line positions while leaving the collision strengths unchanged. Reliable matching requires consistent configurations, term labels and J values, with explicit treatment of repeated or ambiguous labels. Where needed, the split structure is constructed before the energy shifts are applied. Refining these positions does not itself supply resolved-J collisional dynamics.

6.6.2 Statistical splitting of LS multiplets

In LS coupling, the relative strengths of a spin-allowed electric-dipole multiplet follow the angular factor

$$W_{J_u J_\ell} \;\propto\; (2J_u+1)(2J_\ell+1)\begin{Bmatrix} L_u & J_u & S \\ J_\ell & L_\ell & 1 \end{Bmatrix}^2 . \tag{6.16}$$

Normalization over the allowed components gives $\mathrm{PEC}_{J_u J_\ell} = \mathrm{PEC}_{\mathrm{LS}} W / \sum W$. This redistribution assumes LS angular strengths and statistical sublevel populations. split_pec_multiplet evaluates the equivalent Condon–Shortley factors for spin-conserving dipole multiplets with $|\Delta L| \le 1$. Intercombination and $|\Delta L| > 1$ lines remain unsplit at the wavelength between statistically weighted term centers. Configuration mixing and resolved-J population kinetics require additional atomic information.

6.6.3 Doppler and natural profiles

For an emitter of mass m at temperature T_i,

$$\Delta\nu_D = \frac{\nu_0}{c}\sqrt{\frac{2eT_i}{m}}, \qquad \phi_D(\nu) = \frac{\exp[-(\nu-\nu_0)^2/\Delta\nu_D^2]}{\sqrt{\pi}\Delta\nu_D}. \tag{6.17}$$

This is a Gaussian with standard deviation $\Delta\nu_D/\sqrt{2}$ and FWHM $2\sqrt{\ln 2}\,\Delta\nu_D$. The broadening API takes T_i in eV and the emitter mass in kg.

For lifetime broadening of an upper and lower state,

$$\Delta\nu_L = \frac{\Gamma_u + \Gamma_\ell}{2\pi}, \qquad \gamma = \frac{\Delta\nu_L}{2}, \qquad \phi_L(\nu) = \frac{\gamma/\pi}{(\nu-\nu_0)^2+\gamma^2}. \tag{6.18}$$

The Lorentzian FWHM is $\Delta\nu_L$. Density-dependent pressure broadening requires additional physics; it is not supplied merely by this lifetime width.

A change from frequency to wavelength coordinates preserves integrated emissivity through $\phi_\lambda = \phi_\nu |\mathrm{d}\nu/\mathrm{d}\lambda|$, with consistent length units. The implemented profiles evaluate a frequency-shaped kernel on a wavelength grid and normalize its area over that finite grid. This provides a narrow-line representation; an exact transformation of a broad profile would also require the Jacobian. Grid extent and resolution determine the treatment of wings and the numerical normalization.

Profile grid resolution. Natural profiles are evaluated on offsets $\Delta\lambda$ from the line centre, stored in `dwave_arr`, with the frequency offsets $\Delta\nu = -c\,\Delta\lambda/[\lambda_0(\lambda_0+\Delta\lambda)]$. A natural width can be far below the float64 resolution of the wavelength itself (5×10^{-15} nm for the Be I line at 454.8 nm, where float64 resolves 6×10^{-14} nm), and frequencies near 10^{15} Hz are resolved only to 0.1 Hz, so neither absolute grid can carry such profiles. For these lines the absolute wavelengths `wave_arr`, centre plus offset, all round to the same value; their profiles must be read against `dwave_arr`.

6.7 Electron energy loss, radiated power and emission functions

The electron-energy budget includes both emitted radiation and changes in atomic internal energy. For an electron-driven GCR model, these contributions combine as (Summers et al., Eq. 27)

$$L_e = \sum_X \mathcal{U}_X \,\dot{N}_X\big|_{\mathrm{atomic}} + n_\mathrm{e}\sum_\rho \mathrm{PLT}_\rho N_\rho + n_\mathrm{e}\sum_p \mathrm{PRB}_p N_p, \tag{6.19}$$

where X ranges over the retained metastables of the ion and its parent. The energies $\mathcal{U}_X$ are in joules relative to one reference shared across charge stages, and $\dot{N}_X|_{\text{atomic}}$ follows from GCR redistribution, ionization and recombination. PLT includes radiation between retained metastables through Eq. (6.7). Particle conservation makes the choice of common energy zero immaterial.

At ionization equilibrium the internal-energy term vanishes, so the electron energy loss equals the complete radiated power. For an element, the equilibrium power coefficient is (Summers et al., Eq. 53)

$$P^{\text{tot}} = \sum_z \sum_\sigma (\text{PLT}_\sigma^{[z]} + \text{PRB}_\sigma^{[z]}) \Big(\frac{N_\sigma^{[z]}}{N^{\text{tot}}} \Big)_{\text{eq}}. \tag{6.20}$$

This coefficient is in W cm^3 per electron and per ion of the element. The notation $\text{PRB}_\sigma^{[z]}$ refers to recombination power driven by level σ of stage z, emitted by stage $z-1$. The same equilibrium fractions weight individual line coefficients to give the GTN emission function (Summers et al., Eq. 54), the finite-density counterpart of the solar-physics $G(T_e)$:

$$\text{GTN}_{u\ell}^{[z]} = \sum_\sigma \text{PEC}_{u\ell,\sigma}^{[z]\,\text{exc}} \Big(\frac{N_\sigma^{[z]}}{N^{\text{tot}}} \Big)_{\text{eq}} + \sum_p \text{PEC}_{u\ell,p}^{[z]\,\text{rec}} \Big(\frac{N_p^{[z+1]}}{N^{\text{tot}}} \Big)_{\text{eq}}. \tag{6.21}$$

ColRadPy supplies the component coefficients and the retained equilibrium fractions through ionization_balance (`processed['pops_ss']`); the combined energy-loss, element-power and GTN expressions are not assembled automatically. With population rates in cm^{-3} s^{-1}, L_e is in W cm^{-3}. Tabulated wavenumber energies convert to joules as $\mathcal{U} = hc\,(100 E_{\text{cm}^{-1}})$ using SI constants, and the ground energies of successive ions include the intervening ionization thresholds.

The normalization N^{tot} must be consistent with the intended observable. A retained-population sum gives power per retained particle, whereas power per ion of the element requires the full reconstruction in Section 4.4. These expressions omit CX and assume complete radiative channels. The restricted `prb` output therefore provides only part of an electron cooling rate.

7 Projection of the high n-shells

A finite low-level model can omit important capture and cascade pathways through high Rydberg shells, as well as excitation-mediated ionization through those shells. Projection incorporates their effect through an auxiliary bundle-nS model while retaining a compact explicit level system. It is the same quasistatic elimination used in Section 4, applied to shell populations.

The construction has three stages: elimination of the auxiliary high shells, subtraction of direct processes already present in the ADF04 model, and expansion of the remaining corrections onto explicit levels. This separation preserves the low-level atomic data while adding pathways represented by the high-shell calculation.

7.1 Condensed shell model and elimination

For a fixed parent p and spin system, let x_n denote the population of shell n. ADF17 contains condensed bundle-nS couplings calculated by ADAS204 on a temperature–density grid. In its loss-positive convention, the shell balance is

$$\sum_m P_{nm} x_m = n_e N_p r_n. \tag{7.1}$$

The diagonal $P_{nn} > 0$ contains the shell loss rate, and off-diagonal P_{nm} are negative transfer terms from m to n, in s^{-1}. The coefficient r_n describes capture into shell n in cm^3/s. Ionization

from n to parent q is specified by κ_{nq} in s^{-1}; X_q describes parent transfer along paths that bypass the retained shells, in cm^3/s. Parent and spin labels on the arrays are suppressed here.

Because this auxiliary model is evaluated at each density, it supplies the density-dependent redistribution of high doubly excited states that is absent from zero-density ADF04 recombination data (Summers et al., Eq. 45) [9].

Partitioning the shells into retained shells l and higher shells u, solution of the u equations and substitution into the retained balance gives

$$P' = P_{ll} - P_{lu}P_{uu}^{-1}P_{ul}, \qquad r' = r_l - P_{lu}P_{uu}^{-1}r_u, \tag{7.2}$$

$$\kappa' = \kappa_l - \left(P_{uu}^{-1}P_{ul}\right)^{\mathsf{T}}\kappa_u, \qquad X' = X + \kappa_u^{\mathsf{T}}P_{uu}^{-1}r_u. \tag{7.3}$$

These terms account for return from the eliminated manifold to a retained shell and for transfer to a parent. Their signs follow the loss-positive definition of P, which reverses the generator sign convention used for C.

7.2 Removing the direct processes

The reduced auxiliary matrices contain direct low-level processes as well as the desired high-shell pathways. Their direct components must be subtracted before addition to the ADF04 model. ADF17 provides D_{nm} for direct shell coupling, D^{κ}_{nq} for direct ionization and d_n for direct radiative, dielectronic, three-body and charge-exchange recombination. Restricting D to l and removing its diagonal losses to u defines D', since transfers through u now form part of the indirect contribution. The required corrections are

$$P^{\mathrm{ind}} = P' - D' - \mathrm{diag}\left(\sum_q D^{\kappa}_{nq}\right), \qquad \kappa^{\mathrm{ind}} = \kappa' - D^{\kappa}, \qquad r^{\mathrm{ind}} = r' - d, \qquad X^{\mathrm{ind}} = X'. \tag{7.4}$$

The off-diagonal entries of $-P^{\mathrm{ind}}$ are indirect shell gains. The corresponding ionization and recombination differences retain the additional pathways supplied by the auxiliary model.

7.3 Expansion onto the explicit levels

An ADF18 `a17_p208` file associates each explicit level with a parent, spin system and shell $n(i)$. Its expansion weight w_i is normalized over the mapped levels of that shell. Incoming shell population is distributed according to these weights, while an outgoing rate per shell population applies to each constituent level. For one parent and spin group the expansion is

$$R^{\mathrm{ind}}_{ip} = w_i r^{\mathrm{ind}}_{n(i)}, \qquad S^{\mathrm{ind}}_{ip} = \kappa^{\mathrm{ind}}_{n(i)p}/n_{\mathrm{e}}, \tag{7.5}$$

$$\Delta C_{ij} = -w_i P^{\mathrm{ind}}_{n(i)n(j)} \quad (n(i) \neq n(j)), \qquad X^{\mathrm{ind}}_{qp} = \sum_{\text{spin systems of } p} X^{\mathrm{ind}}_{q}. \tag{7.6}$$

The transfer correction is denoted by ΔC to distinguish it from the recombination response G. Transfers within a single shell are omitted because shell populations do not resolve that redistribution.

The expanded corrections enter C_{ij}, C_{pi} and C_{ip} as ΔC_{ij}, $n_{\mathrm{e}}S^{\mathrm{ind}}_{ip}$ and $n_{\mathrm{e}}R^{\mathrm{ind}}_{ip}$, respectively. Matching source-diagonal losses maintain particle conservation. The parent contribution X^{ind}_{qp} augments XCD for $p \to q$. The resulting populations, emission and effective coefficients include high-shell pathways, although the eliminated shell populations and radiation are not explicit outputs.

Parent attribution and mapping. Only ionization to the parent of the shell group is credited. Transfers through core-excited shells into other parents can overlap the excitation-autoionization already present in the ADF04 ionization data. The implementation omits both those gains and their losses, and builds the diagonal from the expanded gains instead of using P^{ind}_{nn} directly. The ADF18 level numbers refer to the original ADF04 order before energy sorting. Unmapped shells/spin systems receive no correction and give a warning. Fractional-parentage switches `IEFPRE`, `IEFPRS` in the older `cbnm93` files and indirect CX are not implemented.

7.4 Grid continuation and numerical qualifications

Interpolation of indirect couplings uses logarithmic temperature, density and coefficient coordinates. Negative inferred gains or ionization, recombination and parent-transfer coefficients are clipped when their magnitude is no greater than 10^{-3} of the projected/direct subtraction scale. Larger negative values cause the affected temperatures to be excluded for that coupling, parent and spin system. This numerical tolerance is independent of any accuracy inferred from the precision of the ADF17 fields.

Below the first retained temperature T_0, a coupling is continued as

$$c(T) = c(T_0)\left(\frac{T}{T_0}\right)^{a} \exp\left[-\Delta E\left(\frac{1}{T}-\frac{1}{T_0}\right)\right]. \tag{7.7}$$

Here $T, T_0, \Delta E$ are in eV. The exponent a is the logarithmic slope of $c\exp(\Delta E/T)$ between the two lowest usable temperatures, restricted to $[-1.5, 0]$ for recombination and parent transfer and to $a \geq 0$ for reduced ionization and shell transfers. Estimated thresholds are the level binding energy χ_i for ionization, the upward parent gap for parent transfer and zero for recombination. A transfer out of level j uses $\chi_j - \bar{\chi}_{n_1} n_1^2/(n_1+1)^2$, based on the mean binding energy of the highest retained shell n_1. Values above the temperature grid or outside the density grid are held at the edge. Warnings identify these continuation regimes, whose effect requires sensitivity assessment.

Routines and data. `read_adf17` reads the condensed model and `read_adf18` its level mapping. `colradpy(..., projection=(adf17, adf18))` invokes `make_projection` and `projection.project`, stores `rates['projection']`, and adds its corrections in `populate_cr_matrix`. Files are read once; `solve_cr` recalculates the projection on each solve's grid. An ionization balance takes one pair or `None` per ADF04 stage.

8 Atomic-data preparation

8.1 Interpolation and asymptotic continuation

Interpolation and continuation complete the mapping from tabulated atomic data to the plasma grid. The coordinate choice is explicit in the interpolation option: a leading `log` or `loglog` qualifier is followed by a SciPy interpolation kind. With `log`, temperature is logarithmic while coefficient values remain linear:

$$\widehat{r}(T_{\mathrm{e}}) = \mathcal{I}[\ln T_k, r_k]\,(\ln T_{\mathrm{e}}) \tag{8.1}$$

The ordinates r_k are unchanged by the coordinate transformation, so linear extrapolation can cross zero and is clipped there. In the Be I data, recombination and ionization coefficients can reach this boundary at temperatures a few times the upper tabulated limit.

By default (`loglog_slinear`), recombination and reduced ionization coefficients are interpolated as

$$\widehat{r}(T_\mathrm{e}) = \exp \mathcal{I}[\ln T_k, \ln r_k]\,(\ln T_\mathrm{e}), \tag{8.2}$$

using the positive values of each transition. At the midpoint in $\ln T$, the result is the geometric mean of the neighboring rates; linear-ordinate interpolation on the same coordinate gives the arithmetic mean. Their ordering alone says nothing about the true coefficient. Log-log interpolation represents a local power law and preserves positivity, but its accuracy remains dependent on the underlying atomic data.

Outside the grid, the two nearest values determine a power-law continuation $\widehat{r} \propto T^s$, subject to prescribed slope limits. Recombination uses $-3/2 \le s \le 0$; reduced ionization uses $-1 \le s \le 1/2$ above the grid and $s \ge 0$ below it, holding a decreasing reduced coefficient constant at the low-temperature edge. Options without `loglog` interpolate coefficient values directly. Excitation and charge exchange use these options, with excitation continued by the Burgess–Tully procedure described below.

The slope restrictions define a continuation policy rather than a general atomic asymptote. For example, $\alpha_\mathrm{DR} \propto T^{-3/2} \exp(-E_r/T)$ has logarithmic slope $-3/2 + E_r/T$ and increases below $T = 2E_r/3$ when both energies are in eV. Monotone continuation can therefore miss a resonance-related rise. Calculations outside the atomic-data grid require additional data or assessment of their extrapolation sensitivity.

Suspected placeholder filtering. Suspected placeholders are screened using thresholds of 10^{-6} cm^3/s for reduced ionization and 10^{-7} cm^3/s for recombination. Leading low-temperature points are also excluded if their decrease toward a later point is steeper than T^{-1} for reduced ionization or T^{-2} for recombination, up to the first acceptable point. These criteria target artifacts in the supported files and can reject valid data outside that context. Warnings identify the affected channels for comparison with the original data; tests are listed in Appendix E.

The Burgess–Tully helpers compress an unbounded temperature coordinate into $0 \le x < 1$. With $\theta = k_B T/\Delta E$ and a positive scaling parameter $c > 1$ (default 1.5), the four implemented transforms are An infinite-energy point, where available, constrains the continuation in transformed

Type	$x(\theta)$	Transformed ordinate y
1	$1 - \ln c/\ln(\theta + c)$	$\Upsilon/\ln(\theta + \mathrm{e})$
2	$\theta/(\theta + c)$	Υ
3	$\theta/(\theta + c)$	$(1 + \theta)\Upsilon$
4	$1 - \ln c/\ln(\theta + c)$	$\Upsilon/\ln(\theta + c)$

Table 5: Coordinate and ordinate definitions in `burgess_tully_rates.py`. The original routines use Kelvin and wavenumbers to form θ.

coordinates. Otherwise the final calculated points determine the continuation, followed by inverse transformation to recover Υ. This permits evaluation beyond a finite data grid: the published Mo I example, whose collision data end near 17 eV, uses such an extension [1]. Agreement with an asymptotic constraint does not establish accuracy far outside the computed range.

Continuation. Above the file grid each transition is continued in the coordinates of its own Burgess–Tully type: along the line from its last calculated point to the infinite-energy point, or, where that point is missing or unusable, along the line through its last two calculated points. Below the file grid it applies a nonnegative transformed continuation or keeps the first collision strength for transitions whose data start far above threshold. This numerical policy prevents negative rates but does not establish atomic-data accuracy outside the calculated range.

8.2 Atomic data from the Flexible Atomic Code

Cross-section data provide an alternative to tabulated thermal rate coefficients. FAC [13] supplies these data together with level energies and radiative rates in a jj-coupled description. read_fac converts its ASCII `PrintTable` outputs to thermal coefficients and the internal dictionary used by ADF04 input. The resulting kinetic calculation then follows the matrix construction and reduction developed in Sections 3–5.

Energy ordering determines the bound-level indices, and levels of the next ion define the parents, with thresholds measured from the lower ion's ground. The interface retains definite J values and weights $2J+1$; unassigned LS labels are stored as $L, S = -1$. Parent detailed balance therefore uses $g_p = 2J_p + 1$.

8.2.1 Common thermal average

For incident kinetic energy E, threshold E_0 and a normalized electron-energy probability density f in eV^{-1}, the thermal coefficient can be written as

$$q(T) = \int_{E_0}^{\infty} \sigma(E) v(E) f(E,T)\,\mathrm{d}E = e^{-E_0/T} T \int_0^{\infty} \sigma(E) v(E) \widetilde{f}(E,T) e^{-u}\,\mathrm{d}u, \quad E = E_0 + uT, \tag{8.3}$$

where $\widetilde{f} = f e^{E/T}$ and T, E, E_0 are in eV. Cross-sections in cm^2 and speeds in cm/s give q in cm^3/s. By default the distribution and speed are relativistic:

$$f(E,T) = \frac{\gamma^2 \beta e^{-\gamma/\theta}}{\theta K_2(1/\theta) m_e c^2}, \qquad \gamma = 1 + \frac{E}{m_e c^2}, \quad \theta = \frac{T}{m_e c^2}, \quad v = c\beta, \tag{8.4}$$

with $\beta = \sqrt{1-\gamma^{-2}}$, K_2 the modified Bessel function, and $m_e c^2$ expressed in eV. The nonrelativistic limit is $f = 2\sqrt{E/\pi} T^{-3/2} e^{-E/T}$. The option `relativistic=False` selects that distribution and the nonrelativistic speed.

The threshold substitution isolates the Boltzmann factor and leaves the reduced coefficient $\widetilde{q} = q e^{E_0/T}$ for evaluation by 64-node Gauss–Laguerre quadrature. A scaled Bessel normalization avoids underflow in the thermal distribution at low temperature. The final physical rate can nevertheless underflow when its threshold factor is sufficiently small.

8.2.2 Excitation and ionization

Excitation and ionization strengths in FAC follow different cross-section conventions. With incident energy E_{H} in hartree, dimensionless $k^2 = 2E_{\mathrm{H}}(1 + \alpha_{\mathrm{f}}^2 E_{\mathrm{H}}/2)$ and a_0 in cm, the corresponding cross-sections are

$$\sigma_{\mathrm{exc}} = \frac{\pi a_0^2 \Omega_{\mathrm{exc}}}{g_\ell k^2}, \qquad \sigma_{\mathrm{ion}} = \frac{a_0^2 \Omega_{\mathrm{ion}}}{g_i k^2}. \tag{8.5}$$

The energy supplied to these formulas must be converted from the eV coordinate used in Eq. (8.3).

The excitation average is stored in the effective-strength convention of Eq. (2.2):

$$\Upsilon_{\ell u} = \frac{\widetilde{q}_{\ell u} g_\ell}{2.1716 \times 10^{-8}} \sqrt{\frac{T}{13.6058}}, \qquad E_0 = \Delta E. \tag{8.6}$$

This representation allows the matrix builder to recover the integrated excitation rate. Ionization is stored in the reduced form $\widetilde{S}$, using the relevant parent threshold as E_0.

Between tabulated energies, strengths are interpolated log-log. Excitation is held at the first value below the grid and joined to the Bethe–Born form $\Omega = b \ln(E/E_0) + c$ above the grid when

that limit is available. Exchange transitions without a Born limit follow the end-point power law, constrained to be at least as steep as E^{-1}. Ionization outside its tabulated range uses FAC's fit,

$$\Omega = p_0 \ln x + p_1 y^2 + p_2 y/x + p_3 y/x^2, \qquad x = E/I, \quad y = 1 - 1/x, \tag{8.7}$$

with high-energy normalization matched to the last tabulated value. These forms specify the continuation of the imported cross-sections used in the thermal average.

8.2.3 Radiative recombination

For electron energy ε and photon energy E_γ, both in hartree, let $\mathcal{B}$ denote FAC's bound-free oscillator-strength density (its tabulated S). The implemented conversion is

$$\sigma_{\mathrm{RR}} = \frac{2\pi\alpha_{\mathrm{f}}\mathcal{B}a_0^2(\alpha_{\mathrm{f}}E_\gamma)^2}{2\varepsilon(1+\alpha_{\mathrm{f}}^2\varepsilon/2)g_p}. \tag{8.8}$$

The recombination average has threshold $E_0 = 0$. Bound-free strength is interpolated log-log, held below the first point, and continued above the grid with FAC's angular-momentum-dependent fit normalized to the final tabulated value. The resulting coefficients enter the recombination term R in Eq. (2.5).

8.2.4 Dielectronic capture and autoionization folding

Capture into an autoionizing level d from parent p is resonant at $E_r = E_d - I_p$, here in eV. If FAC gives resonance strength S_{DC} in $\mathrm{cm}^2\,\mathrm{eV}$, then $\alpha_{\mathrm{DC}} = S_{\mathrm{DC}} v(E_r) f(E_r, T)$. In the nonrelativistic thermal limit its relation to the autoionization rate A^a_{dp} is

$$\alpha_{\mathrm{DC}} = 10^6 \frac{g_d}{2g_p}\left(\frac{h^2}{2\pi m_e e T}\right)^{3/2} A^a_{dp} e^{-E_r/T} \quad (\mathrm{cm}^3/\mathrm{s}), \tag{8.9}$$

using SI h, m_e, e and T in eV. The factor 10^6 converts the translational volume from m^3 to cm^3.

Although ADF04 has fields for Auger rates (`AAUG`) [3], the current main generator represents autoionization through modified rates rather than an explicit decay channel. FAC import therefore folds stabilization and return to parents into the other processes. Let $A^r_{d\ell}$ denote radiative decay rates, $A^r_d = \sum_\ell A^r_{d\ell}$ their sum, and $A^a_d = \sum_p A^a_{dp}$ the total autoionization rate. Stabilization probabilities b_d and return probabilities P_{dp} are evaluated in increasing level-energy order, so downward cascades use the already determined lower-level probabilities:

$$b_d = \frac{\sum_\ell A^r_{d\ell} b_\ell}{A^r_d + A^a_d}, \qquad P_{dp} = \frac{A^a_{dp} + \sum_\ell A^r_{d\ell} P_{\ell p}}{A^r_d + A^a_d}. \tag{8.10}$$

Stable levels have $b = 1$ and $P = 0$. Capture and excitation into an autoionizing level are weighted by its stabilization probability; cascades into an autoionizing lower level carry that lower level's weight. The excitation fraction returning to a parent contributes to excitation-autoionization through

$$\widetilde{S}_{\ell p} \mathrel{+}= \widetilde{q}_{\ell d} P_{dp} e^{-(E_d - I_p)/T}. \tag{8.11}$$

The folding is applicable when collisional rates out of autoionizing levels are small compared with their total decay rates. It represents the tested stabilization pathways in that limit. Capture followed by autoionization into another parent, and radiation emitted before nonstabilizing decay, are not recovered as separate outputs. Nonstabilizing levels are excluded from the folded level

model. Two-photon rates contribute population losses, with the spectral qualification described in Section 6.2.

Tests and resolution limits. read_fac and _fold_autoionization are checked with synthetic tables, nonrelativistic analytic averages, and an explicit autoionization matrix. The folding comparison asserts relative tolerance 10^{-6} on its toy grid without three-body recombination; it does not bound errors for other ions or densities. Paths to these tests are in Appendix E. Magnetic-sublevel tables are not read, and FAC parents have no LS term for the current inter-stage matching. Thermal inverse-rate relations cannot be applied unchanged to an arbitrary nonthermal distribution.

Relativistic equilibrium consistency. FAC forward rates can use Maxwell–Jüttner averages, while three-body inverse rates still use the nonrelativistic factor in Eq. (2.7). For classical free electrons the relativistic/nonrelativistic ratio of translational partition densities is

$$\mathcal{R}(\theta) = \sqrt{\frac{2}{\pi\theta}}\, e^{1/\theta} K_2(1/\theta), \qquad \theta = T_e/(m_e c^2).$$

With the ion partition functions fixed, relativistic detailed balance would multiply the three-body/ionization ratio by $1/\mathcal{R}$. The ratio tends to one at low temperature and is about 1.037 at 10 keV and 1.333 at 85 keV. The present combination does not establish relativistic Saha equilibrium; forward-average checks and nonrelativistic LTE checks test different limits.

9 Charge-state kinetic-energy evolution

The charge-state model can also transport the kinetic energy of a trace species interacting with a prescribed background. Once the charge-state fractions $x_z(t)$ are known, the energy associated with stage z, per modeled particle, is defined as

$$\mathcal{E}_z = \frac{3}{2} x_z T_z. \tag{9.1}$$

Ionization and recombination transfer this energy with the same population generator. Exchange with background ions at temperature T_b adds a relaxation term, giving

$$\dot{\mathcal{E}}_z = \sum_{z'} M_{zz'} \mathcal{E}_{z'} - \frac{2}{3} \nu_z \mathcal{E}_z + \nu_z x_z T_b. \tag{9.2}$$

The coefficient ν_z is defined by $\frac{3}{2} x_z \dot{T}_z = -\nu_z x_z (T_z - T_b)$ at fixed population, so the temperature-relaxation frequency is $2\nu_z/3$. In this convention $\mathcal{E}_z$ is in eV per modeled particle, T_z, T_b are in eV, and M, ν_z are in s^{-1}. Multiplication by e converts the energy to joules. The extension describes kinetic-temperature exchange with a fixed bath; a complete plasma energy equation would also include electron heating, radiation and ionization-potential energy (Section 6.7).

Required charge-state layout. The energy-transfer mapping assigns neutral scattering to entry 0 and Coulomb charges $1, 2, \ldots$ to subsequent entries. Its supported layout therefore has one population per consecutive charge stage, beginning with the neutral, with initial abundances and temperatures in the same order. Although the ionization-balance model supports more general retained manifolds, the energy wrapper has no mapping for several metastables sharing a charge or for an initial charge offset. Those layouts require an extension of the energy model.

9.1 Charged and neutral energy transfer

The charged helper implements

$$v_s^2 = 2eT_s/m_s, \tag{9.3}$$

$$\nu_{ab} = \frac{n_b Z_a^2 Z_b^2 e^4 \ln\Lambda}{\pi^{3/2}\epsilon_0^2 m_a m_b (v_a^2 + v_b^2)^{3/2}}. \tag{9.4}$$

In these expressions temperatures are numeric values in eV, eT is an SI energy, masses are in kg, and n_b is in m^{-3}. The helper evaluates

$$\lambda_D^{-2} = \sum_s \frac{n_s Z_s^2 e^2}{\epsilon_0 e T_s}, \tag{9.5}$$

$$b_{\mathrm{cl}} = \frac{|Z_a Z_b| e^2}{4\pi\epsilon_0 \mu (v_a^2 + v_b^2)/2}, \qquad b_{\mathrm{q}} = \frac{\hbar}{\mu\sqrt{v_a^2 + v_b^2}}, \tag{9.6}$$

$$\ln\Lambda = \ln\frac{\lambda_D}{\max(b_{\mathrm{cl}}, b_{\mathrm{q}})}, \qquad \mu = \frac{m_a m_b}{m_a + m_b}. \tag{9.7}$$

The screened, weakly coupled collision model is not reliable when its assumptions fail or its logarithm ceases to be physically meaningful.

For the neutral, the implemented polarization-scattering approximation uses

$$K_{\mathrm{mt}} = 2.21\pi\alpha_{\mathrm{f}} c\sqrt{\frac{m_e}{\mu}\, a_{\mathrm{pol}} a_0^4}, \qquad \nu_{nb} = \frac{3 n_b \mu}{m_n + m_b} K_{\mathrm{mt}}. \tag{9.8}$$

Here a_{pol} is the numeric polarizability in Bohr-radius cubed and α_{f} is the fine-structure constant. This elastic approximation does not include every long-range or inelastic energy-transfer mechanism.

Energy extension. The energy solver uses one plasma temperature/density point, converts density from cm^{-3} to m^{-3} for the Debye calculation, reconstructs populations from the stored eigenpairs and integrates Eq. (9.2) with BDF. If integration fails, a warning reports the solver message and all stored histories terminate at the last reached time.

10 Applicability and verification

The validity of a collisional-radiative calculation depends jointly on the plasma assumptions, the completeness of the atomic model and the numerical solution of its balance equations. The preceding derivations identify the assumptions associated with each reduction and observable; the principal qualifications are collected here.

The basic model assumes a prescribed electron bath and optically thin emission. Quasistatic reduction additionally requires separated relaxation times, LS splitting requires the specified coupling and sublevel populations, and kinetic-energy evolution assumes a trace species in a fixed background. Relativistic FAC forward rates have the equilibrium qualification discussed in Section 8.2. Approximate completion and extrapolation increase data coverage, while introducing model choices whose effect must be assessed for the application.

10.1 Current limitations

Area	Current qualification	Discussed in
Projection	The fractional-parentage switches of the `cbnm93` ADF17 files and indirect charge exchange are not implemented. Below the ADF17 temperature grid, couplings use estimated thresholds and limited power laws; above it, edge values are held. These continuations and rejection tolerances need sensitivity checks.	Section 7
Radiated power	`prb` is the power of the explicit levels after capture; it omits the recombination continuum, bremsstrahlung and projected-shell radiation in ADAS PRB. Two-photon contributions have correct integrated energy but are not continuum spectra in the generic line outputs.	Sections 6.2, C
Population convention	Stored excitation responses may be normalized; resolved balance fractions refer to retained populations unless the excited populations are reconstructed.	Sections 4.3, 4.4
Relativistic rates	Maxwell–Jüttner forward averages are combined with nonrelativistic three-body detailed balance; relativistic Saha consistency is not established.	Section 8.2
Energy layout	One entry per consecutive charge stage beginning with the neutral is required by the energy-transfer mapping.	Section 9
Processes not modelled	Ionization from the stage below into the metastables (the ADF04 I lines; the ionization from the stage below of the GCR equations of Summers et al.), proton and other ion impact (P lines), and unresolved dielectronic satellite lines and the autoionization of levels of the ADF04 file (L lines and the AAUG values) are not modelled; `read_adf04` skips the P, I and L lines with a warning. The electron energy loss, the radiated power of an element and GTN functions are not assembled.	Sections 5, 6.7
Coupling assumptions	Intercombination and $\lvert\Delta L\rvert > 1$ lines are kept unsplit rather than split with a coupling model.	Section 6.6

Verification begins with coefficient signs, state and parent mapping, column-sum conservation, density powers, population normalization, line-energy consistency and exported units. Transient calculations can be compared with an independent matrix exponential at representative grid points, including the initial state. Quasistatic populations can be substituted back into the excited balance equations. These identities test numerical and bookkeeping consistency independently of the accuracy of the input atomic data.

10.2 Analytic examples and checks

The two-level example illustrates density dependence analytically, while the chain and direct XCD reduction test population evolution and effective-rate construction. The table reports the checks executed by the accompanying generator; the examples themselves appear beside their derivations in Sections 3 and 5.

Quantity	Development	Reference
Chain: $N_1(1)$	0.367879	0.367879
Slow mode: $N_0(10^{12})$	0.367879	0.367879
Be II: XCD$[1, 0]$	3.8811×10^{-12}	3.8811×10^{-12}

Table 7: Executed checks. The first two rows are dimensionless populations. Both XCD values have units cm^3/s.

The figure script records the branch HEAD, working-tree status, package-module hashes, Python and library versions in `provenance.json`. It also hashes its bundled atomic input, the figure and build scripts, and the manual sources under `check_input_sha256` and `documentation_sha256`. These hashes identify the numerical example and documentation inputs. Grid parameters and numerical results are recorded in `checks.json`. The analytic figures and checks establish specific limits and agreements. They are not a certification of all atomic datasets or an ADAS benchmark.

A Data products and external interfaces

A.1 Array layout

Physical indices precede the thermodynamic axes in the stored arrays. A Cartesian temperature–density grid has trailing shape (N_T, N_n), while paired evaluation has one trailing axis N_{pair} and associates corresponding temperature and density entries. Representative layouts are

$$\begin{aligned} \texttt{pops}: &\ (N_e, N_{\mathrm{driver}}, N_T, N_n), \\ \texttt{pecs}: &\ (N_{\mathrm{line}}, N_{\mathrm{driver}}, N_T, N_n), \\ \texttt{scd}, \texttt{acd}: &\ (N_m, N_p, N_T, N_n), \\ \texttt{qcd}: &\ (N_m, N_m, N_T, N_n), \\ \texttt{xcd}: &\ (N_p, N_p, N_T, N_n), \\ \texttt{rates['projection']['parent_coupling']}: &\ (N_p, N_p, N_T, N_n). \end{aligned}$$

An excited-level axis follows the reduced level map and is therefore distinct from the original bound-level numbering. Driver columns also distinguish excitation, recombination and CX contributions; their indices cannot all be interpreted as ionization parents. The orientations of the effective-coefficient arrays are listed below.

A.2 ADF04, ADF11, ADF13, ADF15 and HDF5

The ADAS formats distinguish level-resolved inputs from effective coefficients. ADF04 provides the atomic model of one ion, whereas ADF11 contains charge-state coefficients on temperature–density grids, produced as described in Appendix C. The ADF11 reader retains logarithmic grids and coefficients; interpolation in the ionization-balance calculation precedes conversion to physical values. File families and block metadata identify the coefficient classes. ADF13

Array	First index	Second index	Rate multiplier
`scd`	source bound metastable	destination parent	n_e
`acd`	destination bound metastable	source parent	n_e
`ccd`	destination bound metastable	source CX parent	n_h
`qcd`	source bound metastable	destination bound metastable	n_e
`xcd`	source parent	destination parent	n_e
`parent_coupling`	destination parent	source parent	n_e

Table 8: Stored coefficient orientation. `parent_coupling` is the projected part X^{ind} of XCD in `rates['projection']`, transposed when it is added to `xcd`. Equal matrix dimensions can conceal an accidental transpose, so checking shapes alone is insufficient.

provides line-specific S/XB ratios and driver information [10]; ColRadPy reads these data and can generate them from SCD and excitation PECs (Section 6.4).

ADF15 records photon emission coefficients with their wavelengths and driver identifiers. Its wavelength unit is the angstrom [11], requiring one conversion from the internal nm representation. Each emitted line–driver pair forms a separately indexed block with process and driver metadata. These conventions determine how downstream readers associate coefficients with physical transitions.

Export and reading. The ADF15 wrapper performs the nm-to-angstrom conversion, declares all line–driver blocks and writes distinct selectors, the ion field (for example `C + 2`) and source member name required by the ADAS reader. The ADF13 writer uses the same identification fields, with `INDM = T` for whole-ion normalization and the driver index otherwise. Its ADAS reader retains three selector digits, limiting readable output to 999 blocks.
Input handling follows the corresponding fixed-field conventions: ADF04 uses 8-character value fields; ADF11 blocks are placed by their class-dependent IPRT, IGRD, JPRT and JGRD labels; each ADF13 block carries its own grids and metastable identifier. ADF11 files are collected before missing QCD/XCD arrays are created, so SCD need not appear first. Bundled reader tests are identified in Appendix E; their coverage concerns the supplied fixtures rather than every possible collection or revision.

HDF5 serialization uses an optional dependency and can include PECs, GCR coefficients and additional arrays. A reproducible saved result also records the atomic-file version and hash, retained levels, units, grid ordering, population convention and any supplementary rate model.

A.3 Input routes and retained manifolds

ionization_balance accepts coefficients derived from ADF04 models or precomputed ADF11/HDF5 data. Since retained manifolds differ between stages, a sequence such as $[[0,1],[0,1,2],]$ requires a ragged representation rather than a rectangular numeric array. Input-file order carries the charge-stage ordering and forms part of the calculation definition. Relevant implementation limits are collected in Section 10.

Constructor defaults. For ADF04, a flat metastable array selects levels in the first stage; later selections use the preceding SCD parent count. A per-stage list may have unequal lengths. Default initial abundances are sized after the GCR stages are loaded, including the final parent stage, and place all population in the first state. ADF11 and HDF5 source solves use the same default-abundance helper after their ionization matrix is constructed.

B Numerical methods for matrix evolution

B.1 Constant sources and modal evaluation

For a diagonalizable generator $C = V\Lambda V^{-1}$, the source-free propagator is $V\mathrm{e}^{\Lambda t}V^{-1}N(0)$. Writing the modal populations as $v = V^{-1}N$ and the transformed source as $c = V^{-1}s$, each component satisfies

$$v_k(t) = v_k(0)\mathrm{e}^{\lambda_k t} + c_k\phi(\lambda_k, t), \qquad \phi(\lambda, t) = \begin{cases} (\mathrm{e}^{\lambda t} - 1)/\lambda, & \lambda \neq 0, \\ t, & \lambda = 0. \end{cases} \tag{B.1}$$

Evaluation with `expm1` avoids cancellation in $\mathrm{e}^{\lambda t} - 1$ for small $|\lambda t|$. The same product sets the significance of a slow mode over the observation interval; a small rate alone is not a reason to discard that mode. A constant source can also be included through an augmented exponential,

$$\begin{pmatrix} N(t) \\ 1 \end{pmatrix} = \exp\left[t \begin{pmatrix} C & s \\ 0 & 0 \end{pmatrix} \right] \begin{pmatrix} N(0) \\ 1 \end{pmatrix}. \tag{B.2}$$

B.2 Complex modes and real populations

A real, nonsymmetric rate matrix can have complex conjugate eigenvalues. For example, a conservative three-state cycle has

$$C = \begin{pmatrix} -1 & 0 & 1 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}, \qquad \lambda = 0,\ -\tfrac{3}{2} \pm \tfrac{\sqrt{3}}{2}\,\mathrm{i}. \tag{B.3}$$

Conjugate modes together produce real populations for real initial conditions and sources. Their imaginary parts cancel in the exact reconstruction, whereas discarding imaginary parts of the individual eigenvalues or eigenvectors changes the propagator. A small imaginary residual in the reconstructed population may be removed only after its numerical significance has been assessed. Real-arithmetic exponential methods avoid these complex intermediates.

B.3 Defective and ill-conditioned eigenbases

The equal-rate chain in Eq. (3.9) illustrates a physical generator without a complete eigenbasis. Its Jordan block produces the te^{-t} term. A numerical eigenvector inverse can yield plausible populations while losing this contribution. Near such matrices, the conditioning of the eigenbasis controls reconstruction accuracy, motivating a stable matrix exponential when the modal representation is unreliable.

Transient algorithms. The transient solvers check eigenvector conditioning and compare selected populations against a matrix exponential. Untrusted points are recomputed with scipy.linalg.expm; a constant source uses an augmented matrix. Real populations are returned, and slow physical modes are retained according to their evolution over the requested times rather than an absolute rate cutoff. The cached evaluator needs the original rate matrix to perform the same exponential fallback.

B.4 Stationary numerical methods

For a particle-conserving matrix with nonnegative off-diagonal rates, the solver first attempts Grassmann–Taksar–Heyman elimination [15]. Its use of off-diagonal rates avoids subtractive

cancellation and can preserve very small stationary populations in suitable communicating systems. Its applicability still depends on representable rates and the elimination conditions; underflow, overflow and nonuniqueness in reducible systems remain possible. When this path is unavailable, the solver tests the smallest-magnitude eigenmode against residual and normalization criteria and replaces unreliable results with a normalized stationary solve. The ionization-balance wrapper uses the same routine.

C How ADAS makes its ADF11 data

ADF11 archives effective coefficients for all charge stages of an element on electron-temperature and density grids [5]. Its unresolved representation has one population per stage and contains ACD, SCD, CCD, PLT, PRB and PRC. The metastable-resolved representation subdivides these classes by retained level and parent and adds QCD and XCD [9]. These latter classes describe redistribution within a stage and consequently have no separate counterpart in a one-population stage model.

C.1 Production of GCR-project coefficients

The GCR-project class `96` is produced through a sequence of calculations. ADAS204 [6] solves a bundle-nS model and supplies the condensed matrices used for high-shell projection. ADAS208 [7] combines that projection with an ADF04 model and partitions the reduced matrix into effective coefficients. Ionization is taken from the file's S lines where available and from ECIP otherwise, with optional replacement of metastable rates by zero-density ADF07 data. Three-body recombination follows by detailed balance, as in ColRadPy.

The coefficients are archived by isoelectronic sequence in ADF10 [4] on scaled grids T_e/z_1^2 and n_e/z_1^7, where z_1 is the charge of the recombining ion. ADAS404 [8] constructs the element-specific ADF11 files by cubic-spline interpolation in charge, temperature and density. PRB follows a different production path: ADAS204 supplies the bundle-n recombination power, including continuum radiation, spin-weighted bremsstrahlung, cascades and dielectronic stabilization.

C.2 XCD in the inspected ADAS208 implementation

The ADAS208 source inspected for this manual is `adas208.for`, version 1.23, dated 28 February 2020. Its parent cross-coupling multiplies ordinary-level ionization coefficients, in cm^3/s, by recombination-driven populations returned by `B8STKE`. That routine documents these populations in cm^3, corresponding to normalization per $n_\mathrm{e}N_p$. The contraction lacks the factor n_e in Eq. (5.4). The ADF04-level contribution is consequently scaled by $1/n_\mathrm{e}$, strongly suppressing it at plasma densities and leaving XCD dominated by the projected parent coupling.

A comparison with the Be^+ ADF10 file `pj#xcd96_li21.dat` is consistent with this behavior in an earlier calculation. The file uses $z_1 = 2$ and was produced in 2006 from `cpb03_ls#be1.dat`. For transfer from the $1s2s\ {}^3S$ parent to $1s^2\ {}^1S$, its XCD agrees with ColRadPy's projected contribution alone to 0.3 % at the tabulated nodes between 8000 and 40000 K and between 1.28×10^{11} and $1.28 \times 10^{14}\ \mathrm{cm}^{-3}$. Including ColRadPy's ordinary-level path gives 1.0–1.9 times the ADF10 value. In this example, dielectronic capture from the metastable parent strongly populates the $1s^2nl$ levels, making the omitted density multiplier consequential. The implementation finding applies to the inspected source; the earlier data comparison is supporting numerical evidence.

C.3 Resolved and unresolved charge-state data

For the GCR-project classes `96` and the earlier `93`, unresolved coefficients are obtained by contraction of the resolved coefficients rather than by an independent ground-only calculation. ADAS404 uses the metastable fractions in Eq. (5.7). Only stages with several retained metastables are affected. In the `96r` collections these include He^0, Li^+, Be^0 and Be^{2+}, and all stages of C, N, O, Ne and Si other than the bare nuclei and the H-, Li- and Na-like stages.

The required MET ratios form an ADF10 class, `adf10/met96/pj#met96_<seq>##.dat`, written by ADAS208. They are ground-relative ratios $\eta_\rho = N_\rho/N_g$; for example, $\eta_{^3P} = 1.49$ for Be^0 at 2 eV. ADAS404 interpolates and normalizes them to metastable fractions before contraction. Other unresolved classes, including `74`, `85`, `89`, `92`, `12` and `50`, originate from other production codes, identified in the file comments, and need not have associated resolved data.

C.4 Resolved and unresolved emission data

For photon emissivity and S/XB data, the unresolved construction uses ground condensation. The ADAS prefixes `pju` and `llu`, with `INDM = T`, identify calculations with the ground as the sole retained level and whole-stage population normalization. Their resolved counterparts, `pjr` and `llr`, retain metastable populations. ColRadPy represents the unresolved construction by retaining only the ground, with the associated normalization described in Section 4.3.

D Source-code map

The map below is generated from the same source as the numerical checks. Paths are relative to the repository root.

File	Entry point	Line
`colradpy/colradpy_class.py`	populate_data	643
same	make_ioniz_from_reduced_ionizrates	818
same	suppliment_with_ecip	895
same	make_cx_rates_from_file	958
same	make_recombination_rates_from_file	1001
same	make_three_body_recombination	1075
same	make_electron_excitation_rates	1120
same	populate_cr_matrix	1282
same	solve_quasi_static	1603
same	solve_time_dependent	2052
same	split_pec_multiplet	2151
same	shift_j_res_energy_to_nist	2568
same	solve_cr	2704
same	make_nat_broad	2746
same	make_dopp_broad	2824
same	write_pecs_adf15	3150
same	get_sxb	3210
same	write_sxb_adf13	3246
`colradpy/ionization_balance_class.py`	populate_ion_matrix	531
same	solve_no_source	679
same	solve_source	717
same	solve_time_independent	782
same	metastable_fractions	44

File	Entry point	Line
same	collapse_metastables	490
colradpy/solve_matrix_exponential.py	solve_matrix_exponential	245
same	eval_matrix_exponential_solution	318
same	solve_matrix_exponential_steady_state	349
same	solve_matrix_exponential_source	442
colradpy/energy_balance.py	solve	88
same	plasma_energy_transfer_frequency	243
same	neutral_energy_transfer_frequency	298
same	coulomb_logarithm	351
same	debye_length	393
colradpy/read_fac.py	read_fac	642
same	_fold_autoionization	595
same	fac_cross_sections	505
colradpy/convolve_eedf.py	rate_coefficients	68
same	resonance_rate_coefficients	105
same	maxwellian	29

E Evidence and reproduction paths

The figure and check generator uses `examples/input/cpb03_ls#be1.dat` for the Be II XCD check and analytic matrices for the chain and slow mode. Its numerical results and provenance record accompany this PDF. Run it from the repository root as described on the documentation page.

The source-controlled tests below address complementary properties of the calculation. Their synthetic or bundled inputs make each comparison independently inspectable.

- Population normalization: excitation versus parent columns and the normalized direct ionization term.
- FAC reader and rates: synthetic tables written by the tests, thermal averages and explicit versus folded autoionization (`FoldingTests.test_explicit_autoionization`, asserted relative tolerance 10^{-6}).
- Matrix solvers and LTE helpers: analytic/exponential comparisons and nonrelativistic equilibrium relations.
- Rate interpolation, placeholder filtering and projection: the implemented interpolation, rejection and continuation policies.
- Reader fixtures in ADF04, ADF11, ADF13, ADF17 and ADF18 tests.

Individual checks can be reproduced from the repository root. For example, the FAC tests are run with

```
python -m pytest tests/test_read_fac.py -q
```

An asserted tolerance specifies acceptance for the stated test and reference; it is not an error bound over physical atomic datasets.

Reproduction of an external benchmark additionally requires the exact script and configuration, atomic-file identifiers and hashes, code revision, population conventions, grids, reference quantities and numerical outputs.

References

[1] C. A. Johnson, S. D. Loch and D. A. Ennis, "ColRadPy: A Python collisional radiative solver," *Nuclear Materials and Energy* (2019). doi:10.1016/j.nme.2019.01.013. The repository includes a six-page author manuscript as `docs/source/files/colradpy_paper.pdf`.

[2] ADAS, *OPEN-ADAS report 1*, especially the generalized collisional radiative definitions and separate emission drivers. https://www.adas.ac.uk/adaseu_site/reports/open_1.pdf.

[3] ADAS User Manual, Appendix A, *ADF04: resolved specific ion data collections*. https://www.adas.ac.uk/man/appxa-04.pdf.

[4] ADAS User Manual, Appendix A, *ADF10: iso-electronic master files*. https://www.adas.ac.uk/man/appxa-10.pdf.

[5] ADAS User Manual, Appendix A, *ADF11: iso-nuclear master files*. https://www.adas.ac.uk/man/appxa-11.pdf.

[6] ADAS User Manual, *ADAS204: specific z recom/ionis – process collisional radiative coefficients and populations*. https://www.adas.ac.uk/man/chap2-04.pdf.

[7] ADAS User Manual, *ADAS208: specific z excitation – advanced population processing*. https://www.adas.ac.uk/man/chap2-08.pdf.

[8] ADAS User Manual, *ADAS404: isonuclear master data – extract from isoelectronic master data*. https://www.adas.ac.uk/man/chap4-04.pdf.

[9] H. P. Summers *et al.*, "Ionization state, excited populations and emission of impurities in dynamic finite density plasmas: I. The generalized collisional-radiative model for light elements," *Plasma Physics and Controlled Fusion* **48**, 263 (2006). doi:10.1088/0741-3335/48/2/007.

[10] ADAS User Manual, Appendix A, *ADF13: ionisation per photon coefficients*. https://www.adas.ac.uk/man/appxa-13.pdf.

[11] ADAS User Manual, Appendix A, *ADF15: photon emissivity coefficients*. https://www.adas.ac.uk/man/appxa-15.pdf.

[12] ColRadPy development branch working tree used for this manual. The recorded HEAD, source-file hashes and runtime versions are in `docs/source/theory/figures/provenance.json`.

[13] M. F. Gu, "The flexible atomic code," *Canadian Journal of Physics* **86**, 675 (2008). doi:10.1139/P07-197; https://github.com/flexible-atomic-code/fac.

[14] G. W. F. Drake, "Spontaneous two-photon decay rates in hydrogenlike and heliumlike ions," *Physical Review A* **34**, 2871 (1986).

[15] W. K. Grassmann, M. I. Taksar and D. P. Heyman, "Regenerative analysis and steady state distributions for Markov chains," *Operations Research* **33**, 1107–1116 (1985).